Option Loss and Transaction Cascades in the Housing Market*

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Abstract

In booming housing markets, transactions may generate their own momentum as property sales create urgency among other buyers. To rationalize this mechanism, we develop a dynamic sequential search model in which property sales signal tighter market conditions and generate transaction cascades. Using data from a major Chinese housing platform, we exploit the quasi-random timing of sales of previously visited properties to identify option-loss shocks. We find that option loss increases buyers' purchase probability by 67%, whereas sales of similar but unvisited properties have little effect, highlighting the importance of inclusion in the buyer's choice set. Consistent with the mechanism, the effect is stronger in tighter markets and among buyers with larger choice sets. A back-of-the-envelope calculation suggests that these cascades account for about 30% of market-level transaction activity. Option loss also reduces subsequent search, broadens search criteria, and is associated with higher transaction prices. Our findings identify option loss as a micro-level mechanism through which individual transactions trigger further transactions and amplify housing market activity.

Keywords: Housing Search; Option Loss; Transaction Cascades; Imperfect Recall
JEL Classification: R31, G41, D83

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1 Introduction

Housing is the largest asset held by many households, making home purchase one of the most consequential financial decisions households face. Yet housing transactions take place in a dynamic market in which buyers compete for a limited set of available properties and attractive houses can disappear quickly. This is particularly salient in booming markets, where frequent transactions may themselves convey information about market tightness. When a property is sold, the transaction not only determines the allocation of that particular house but may also signal to other active buyers that desirable options are becoming harder to secure and that delaying a purchase is increasingly costly. This raises a natural question: can one housing transaction trigger further transactions, generating self-reinforcing transaction momentum in the housing market?

This question matters for understanding why housing market booms may become self-reinforcing. Existing explanations of housing booms often emphasize credit conditions [Favara and Imbs, 2015], belief-driven demand [Piazzesi and Schneider, 2009], and market fundamentals such as housing supply constraints [Glaeser et al., 2008]. This paper highlights a complementary mechanism operating through buyers' search process. When a property is sold, the transaction not only clears one buyer–seller match; it also removes an option from the choice sets of other buyers who had previously considered the same property. By signaling tighter market conditions and lowering the value of continued search, one sale may induce affected buyers to accelerate their own purchases. Housing transactions may therefore generate their own momentum: one transaction triggers additional transactions, creating transaction cascades that amplify market activity.[1]

In addition, this question is theoretically important because it highlights an overlooked search friction: imperfect recall in a dynamic housing market. A growing literature has documented different sources of search frictions in housing markets [Anenberg, 2016, Han and Strange, 2015, Koster and van Ommeren, 2020]. Less is known, however, about imperfect recall generated by property exit.[2] Recall

[1] A large literature has shown that aggregate house price changes are highly positively autocorrelated, a pattern commonly referred to as house price momentum [Case and Shiller, 1989, Guren, 2018]. This price momentum helps explain why housing markets may become self-reinforcing over the cycle. Rather than studying the persistence of price changes, we examine whether individual transactions generate transaction momentum by affecting the purchase decisions of other buyers and thereby triggering further transactions.

[2] Daughety and Reinganum [1992] discuss imperfect recall in search models, but it is not driven by exogenous product exit; instead, it arises from firms' endogenous strategies. Brown and Uru [2024] study generalized Pandora's problems in which alternatives may randomly disappear during search and acquisition is uncertain. This setting is closely related to imperfect recall generated by property exit, although their focus is on optimal and approximately optimal search policies under disappearing alternatives.

is central to sequential search because it allows searchers to return to previously inspected options, thereby shaping when they stop searching and make a purchase [Weitzman, 1979]. In a dynamic market, previously inspected homes may be sold before the searcher decides, making recall infeasible even when buyers remember the property. This friction changes the payoff from continuing search and shapes search intensity, purchase timing, and the properties buyers ultimately choose. This friction is therefore important because it shows how imperfect recall in search can shape aggregate transaction activity and price dynamics.

To understand how property exit can generate transaction cascades in a dynamic market, we formalize a dynamic sequential search model with recall, option loss, and Bayesian updating. The model departs from the canonical Pandora's Box framework of Weitzman [1979], in which previously inspected options remain available and searchers follow a stationary reservation-value rule. In our setting, the sale of a previously inspected property has two effects. It removes an option from the buyer's choice set, making recall imperfect, and it provides information about market tightness. Because buyers do not directly observe market conditions, they use observed sales of previously inspected properties to update their beliefs. A sale shifts beliefs toward tighter market states, lowers the continuation value of search, and induces marginal buyers to stop searching and purchase. The model therefore predicts an immediate increase in purchase probability, a decline in continued search, and higher purchase prices after option loss.

We bring this model to the data by examining how the sale of a property affects other buyers who had previously considered it. We use proprietary data from one of the largest housing listing and transaction platforms in China. A key advantage of the dataset is that it records buyers' on-site property visits together with detailed transaction records, allowing us to identify buyers who had visited a property before it was sold to someone else. We treat these buyers as exposed to an option-loss shock and exploit the quasi-random timing of such sales to estimate their effects on subsequent search behavior and purchase outcomes.

The empirical evidence supports the model's main predictions. First, loss of available options leads to an immediate increase in searchers' purchase probability. The effect is economically large: daily purchase probability rises by about 0.6 percentage points, corresponding to a 67 percent increase relative to the mean among unaffected buyers. The estimate is robust across specifications and varies meaningfully across searcher and property types, including cash versus mortgage buyers and cheaper versus more expensive transacted properties. Importantly, sales of otherwise similar nearby properties that were not previously visited by the buyer have little effect, indicating that prior visitation, and thus entry into the buyer's choice set, is a key channel. The response is also stronger in tighter markets and among buyers with larger choice sets, consistent with a sequential search

prediction. These findings point to a transaction cascade: the sale of one property generates externalities for other searchers by inducing faster subsequent transactions. A back-of-the-envelope quantification suggests that this cascade mechanism amplifies the baseline market transaction probability by a multiplier of 1.43 and accounts for approximately 30 percent of observed market-level transaction activity.

Second, consistent with the model's prediction that option loss reduces the attractiveness of continued search, we find a sharp decline in the number of visits following option loss. This decline is driven primarily by fewer visits to new properties, while recall visits to previously inspected properties decrease only slightly. In subsequent visits, searchers also relax their initial search criteria: they become more likely to consider properties whose prices and floor areas exceed their prespecified preference ranges. Finally, option loss is associated with higher transaction prices. This pattern is consistent with accelerated purchase decisions after option loss: affected searchers may have weaker bargaining positions or become more willing to relax their budget constraints, leading to higher purchase prices.

These results have important implications. Most directly, we provide new evidence on imperfect recall as a search friction in dynamic housing markets, showing that the exit of previously visited properties removes valuable recall options from buyers' choice sets. The resulting contraction of the choice set accelerates purchase decisions, reduces continued search, and leads buyers to broaden their search criteria. Second, the findings imply a redistribution of surplus between buyers and sellers. The dynamic exit of properties appears to weaken buyers' position: affected searchers transact more quickly and pay higher prices. These patterns suggest that sellers may benefit from the urgency created by option loss, especially in tighter markets. Finally, our results provide micro-founded evidence on the amplification of housing market booms. By showing that individual transactions can affect the decisions of other buyers, the cascade mechanism helps explain why transaction activity may become self-reinforcing in a booming market, as observed in the Beijing housing market in our data.

Related Literature. Our paper contributes first to the literature on sequential search. Canonical search models study optimal stopping under costly sampling and characterize search behavior using reservation rules or reservation values [Weitzman, 1979]. A related strand relaxes the perfect-recall benchmark and studies how recall frictions reshape stopping decisions, including uncertain recall [Karni and Schwartz, 1977], endogenous recall [Daughety and Reinganum, 1992], costly revisits [Janssen and Parakhonyak, 2014], and search with deadlines and uncertain recall [Akın and Platt, 2014]. More recently, the literature has also examined environments in which buyers may face frictions in returning to previously sampled options or in securing them after search, such as search deterrence or acquisition

uncertainty [Armstrong and Zhou, 2016, Brown and Uru, 2024]. Within housing, existing work emphasizes search frictions, matching, and market segmentation, showing that housing transactions are shaped by costly search and dynamic market conditions [Fan et al., 2025, Genesove and Han, 2012, Han and Strange, 2015, Piazzesi et al., 2020, Williams, 2018]. Our paper adds to this literature by studying a dynamic housing market in which recall is impaired not by an exogenous memory friction alone, but by the disappearance of previously visited houses. In our setting, the sale of a previously inspected property removes an option from the buyer's effective choice set and also conveys information about market tightness, thereby changing both the value of waiting and the timing of purchase. Empirically, we show that such option-loss shocks increase purchase probability immediately, especially through recall purchases, reduce visits to new houses, and are associated with higher transaction prices; theoretically, we rationalize these patterns in a continuous-time search model with recall, option loss, and Bayesian updating.

Our paper also contributes to the literature on housing-market dynamics. A large literature studies the joint dynamics of house prices, sales, liquidity, and time on market, emphasizing the role of search frictions, credit constraints, and endogenous entry and exit in shaping market outcomes [Ortalo-Magne and Rady, 2006, Stein, 1995, Wheaton, 1990]. More recent work shows that housing-market fluctuations can be amplified by feedback effects and strategic complementarities: shocks to demand or liquidity can propagate through market tightness, transaction speed, listing behavior, and price setting, generating persistence in both transactions and prices [Guren, 2018, Head et al., 2014, Ngai and Sheedy, 2020, 2024, Novy-Marx, 2009]. A related strand emphasizes that congestion and transaction sequencing decisions can themselves move prices and liquidity in frictional housing markets [Grindaker et al., 2026, Moen et al., 2019]. Our paper complements this literature by identifying a new buyer-side propagation mechanism. We show that when a previously visited house is sold to another buyer, this option-loss shock changes subsequent search and purchase behavior: buyers become more likely to transact immediately, especially through recall purchases, reduce search over new properties, relax search criteria, and pay higher prices.

Lastly, our paper also contributes to the behavioral literature on choice deferral and option unavailability. One strand of this literature emphasizes a menu-simplification channel: when decision makers face larger or more conflictual menus, they may be more likely to defer choice, select a default, or choose a no-purchase option, so removing an alternative can sometimes make decision-making easier by reducing comparison difficulty and cognitive burden [Chernev et al., 2015, Dean et al., 2022, Dhar, 1997, Gerasimou, 2018, Tversky and Shafir, 1992]. A related strand, however, emphasizes a signal-based channel: the disappearance of an option can affect behavior not only because it changes the remaining menu, but also

because it conveys information about scarcity, popularity, or high demand. In this literature, sold-out or unavailable products can create urgency, reduce deferral, and increase purchase propensity when unavailability is interpreted as evidence that attractive opportunities are being quickly absorbed [Ge et al., 2009, Steinhart et al., 2013]. To our knowledge, this paper is the first to identify the effect of losing a previously visited house in the housing market.

The remainder of the paper proceeds as follows. Section 2 develops the dynamic search model. Section 3 describes the institutional background and data. Section 4 presents the research design and empirical strategy. Section 5 reports the main results on purchase and search behavior. Section 6 discusses heterogeneity, mechanism evidence, and transaction cascades. Section 7 concludes.

2 Model

This section develops a tractable three-period model that rationalizes the effect of a sale shock on purchase probability through an information-updating channel. Because the buyer does not observe market tightness directly, the sale of a previously inspected house serves as a signal: it shifts her beliefs toward tighter market states, lowers her continuation value, and contracts her stopping threshold. The model is deliberately stripped down so that the stopping threshold and the treatment effect of an observed sale admit closed-form expressions, isolating the signal mechanism in its sharpest form. All proofs for this section are collected in Appendix D. To show that this same mechanism can account for the broader set of empirical patterns we document, including the reduction in visits to new properties, the amplification by choice set size, and the higher transaction prices, we develop a full continuous-time version of the model in Appendix E.

2.1 Setup

There are three dates, $t = 0, 1, 2$. There are two possible market states, loose L and tight T. The true market state remains fixed over the three periods but is unobserved by buyers, who form and update beliefs about the prevailing state over time. At $t = 0$ the buyer holds a prior $\pi_0 \in (0, 1)$ that the market is in the tight state T rather than the loose state L. Sale hazards satisfy $h_T > h_L$, and the expected hazard given any belief π is

$$m(\pi) = \pi h_T + (1 - \pi) h_L. \tag{1}$$

Information between $t = 0$ and $t = 1$. The buyer monitors one previously inspected house y, which sells during $(0, 1]$ with probability h_θ conditional on the

market state $\theta \in \{T, L\}$. The buyer observes whether y sold:

$$S \equiv \{y \text{ sells}\}, \qquad \overline{S} \equiv \{y \text{ does not sell}\}.$$

Choice at $t = 1$. A house with surplus $b \in [0, 1]$ becomes available to the buyer, drawn from $U[0, 1]$ independently of θ. With her belief updated from π_0 to either π_+ (after S) or π_- (after $\overline{S}$), she chooses to stop (purchase house b for payoff b) or continue. If the buyer chose to continue, she pays cost $\kappa \in (0, \frac{1}{2})$.

Choice at $t = 2$. A new house with surplus $x \sim U[0, 1]$ then arrives at $t = 2$. House b remains available with probability $1 - m(\pi)$; otherwise it has been sold in the background. The buyer makes her second decision: she purchases whichever available house has higher value, receiving payoff $\max(b, x)$ if house b survives and x otherwise.

Figure 1 summarizes the timing.

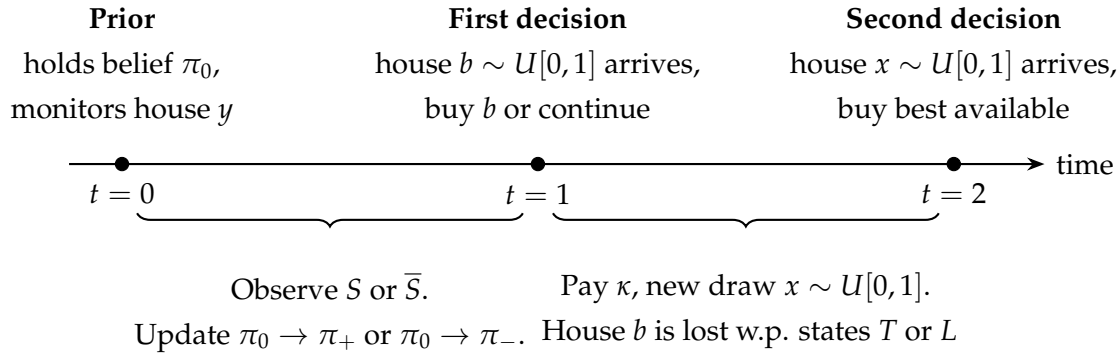


Figure 1: Timeline of the three-period model

Notes: The buyer holds a prior π_0 at $t = 0$ and observes whether the monitored house y sold during $(0, 1]$, updating her beliefs accordingly. At $t = 1$ she draws house b and chooses whether to purchase it or continue. If she continues, she draws house x at $t = 2$ and purchases whichever available house has higher value.

2.2 Bayesian Updating

By Bayes' rule, the buyer's posteriors after observing S and $\overline{S}$ are

$$\pi_+ = \frac{\pi_0 h_T}{\pi_0 h_T + (1 - \pi_0) h_L}, \tag{2}$$

$$\pi_- = \frac{\pi_0 (1 - h_T)}{\pi_0 (1 - h_T) + (1 - \pi_0)(1 - h_L)}. \tag{3}$$

Proposition 1 (Two-sided belief updating). *If $h_T > h_L$ and $\pi_0 \in (0, 1)$, then $\pi_- < \pi_0 < \pi_+$ and correspondingly $m(\pi_-) < m(\pi_0) < m(\pi_+)$.*

Both inequalities follow from the fact that observing a sale is more likely under the tight state ($h_T > h_L$), so observing S raises the posterior on T and observing $\overline{S}$ lowers it. The same ordering carries over to perceived hazards because $m(\pi)$ is increasing in π. The magnitude of the belief update depends on how informative the sale event is, that is, on the gap $h_T - h_L$.

2.3 Value Functions and the Stopping Rule

The payoff from stopping at $t = 1$ is $V^{\text{stop}}(b) = b$: the buyer locks in the recalled option for payoff b. The payoff from continuing, given post-update belief π with implied hazard $m = m(\pi)$, is

$$V^{\text{cont}}(b, m) \;=\; \underbrace{-\kappa}_{\text{search cost}} \;+\; \underbrace{(1-m)\,\mathbb{E}[\max(b, x)]}_{b \text{ survives: pick better of } b \text{ and } x} \;+\; \underbrace{m\,\mathbb{E}[x]}_{b \text{ is lost: keep only } x}. \tag{4}$$

The three terms have distinct economic interpretations. The first term, $-\kappa$, is the flow cost incurred to continue searching to $t = 2$. The second term covers the event in which the recalled house b survives in the background (probability $1 - m$); the buyer then chooses the better of her recalled option b and the new draw x, with expected value $\mathbb{E}[\max(b, x)]$. The third term covers the event in which b has been sold to someone else in the background (probability m); the buyer is then left with only the new draw x, with expected value $\mathbb{E}[x]$. The perceived hazard m therefore governs not the buyer's chance of finding a better house, which depends on the distribution of x, but the chance of losing the fallback option she already has.

For $x \sim U[0, 1]$, $\mathbb{E}[\max(b, x)] = \frac{1}{2} + \frac{b^2}{2}$ and $\mathbb{E}[x] = \frac{1}{2}$, so the value of continuing simplifies to

$$V^{\text{cont}}(b, m) \;=\; -\kappa + \tfrac{1}{2} + \tfrac{(1-m)\,b^2}{2}. \tag{5}$$

Define the *net gain from search*:

$$G(b, m) \;\equiv\; V^{\text{cont}}(b, m) - V^{\text{stop}}(b) \;=\; \tfrac{(1-m)\,b^2}{2} \;-\; b \;+\; \tfrac{1}{2} \;-\; \kappa. \tag{6}$$

The buyer stops at $t = 1$ if and only if $G(b, m) \leq 0$.

Proposition 2 (Threshold stopping rule)**.** *For each $m \in [0, 1]$ there exists a unique $b^*(m) \in (0, 1)$ such that the buyer stops if and only if $b \geq b^*(m)$.*

The result is standard. The net gain $G(b, m)$ is decreasing in b because a higher recalled value b raises the value of stopping faster than it raises the value of continuing. The net gain is positive at $b = 0$, where the buyer prefers to continue and try her luck at $t = 2$, and negative at $b = 1$, where continuing offers no upside but still incurs the cost κ and the risk m of losing the option. Monotonicity and continuity then give a unique threshold $b^*(m)$. A closed-form expression for $b^*(m)$, obtained by solving the quadratic $G(b^*, m) = 0$, is derived in Appendix D.

Proposition 3 (Threshold falls in the perceived hazard). *$b^*(m)$ is strictly decreasing in m on $[0, 1)$.*

As m rises, house b becomes more likely to be lost before $t = 2$ and the expected return from continuing falls, while the value of stopping at b is unchanged. The set of b at which continuation dominates therefore shrinks, and the threshold b^* contracts. Buyers who previously would have preferred to continue now may find stopping optimal.

2.4 Treatment Effect of an Observed Sale

The previous two subsections establish the two building blocks of the signal mechanism: a sale shifts beliefs toward tighter market states (Proposition 1), and a higher perceived hazard contracts the stopping threshold (Proposition 3). Combining these results delivers the model's central empirical prediction: an observed sale raises the buyer's purchase probability at $t = 1$.

Proposition 4 (Observed sales trigger immediate purchase). *Let $b^*_+ \equiv b^*(m(\pi_+))$ and $b^*_- \equiv b^*(m(\pi_-))$. Then $b^*_+ < b^*_-$, and a buyer with $b \in [b^*_+, b^*_-)$ continues searching after observing no sale but stops and purchases at $t = 1$ after observing a sale. Since $b \sim U[0, 1]$, the change in aggregate purchase probability between the two information events is*

$$\Delta P \;=\; b^*_- - b^*_+ \;>\; 0. \tag{7}$$

Figure 2 illustrates the mechanism graphically. The upper curve plots $G(\cdot, m_-)$, the buyer's net gain after observing no sale; the lower curve plots $G(\cdot, m_+)$, the net gain after observing a sale. Each crosses zero at the corresponding stopping threshold.

The thresholds b^*_+ and b^*_- partition buyers into three types. Those with $b < b^*_+$ have weak recalled options and continue regardless of the signal; those with $b \geq b^*_-$ have strong recalled options and purchase regardless of the signal. Only buyers in the middle interval $[b^*_+, b^*_-]$ are responsive: they continue under no-sale beliefs but stop and purchase under post-sale beliefs. This middle interval is the population that generates the treatment effect ΔP in Proposition 4.

Proposition 4 delivers the model's empirical content in closed form. The treatment effect ΔP is unambiguously positive whenever the prior is interior and the two market states are distinguishable ($h_T > h_L$), and its magnitude is governed by how much the sale shock shifts the stopping threshold. This is the theoretical counterpart of the empirical pattern we document in Section 5: a sale shock to the choice set immediately raises the affected buyer's purchase probability.

Appendix E develops the complete version of the model in continuous time. In that environment, previously inspected houses expire as Poisson processes whose

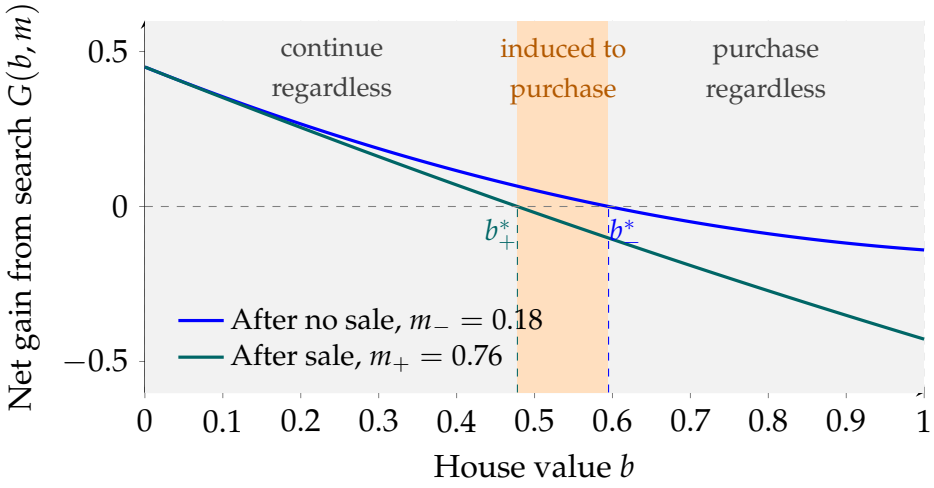


Figure 2: Net gain from search $G(b, m)$ after observing a sale (m_+) and after observing no sale (m_-).

Notes: Parameters: $\pi_0 = 0.5$, $h_T = 0.8$, $h_L = 0.05$, $\kappa = 0.05$, implying $\pi_+ \approx 0.94$ and $\pi_- \approx 0.17$. Observing the sale of the monitored house lowers the curve and shifts the stopping threshold from $b^*_- \approx 0.60$ to $b^*_+ \approx 0.48$. The thresholds partition the b-axis into three regions: buyers with $b < b^*_+$ continue regardless of the signal, buyers with $b \geq b^*_-$ purchase regardless of the signal, and buyers in the orange interval $[b^*_+, b^*_-]$ are induced to purchase by observing the sale.

intensity depends on an unobserved market-tightness parameter θ, and new listings arrive continuously. The buyer monitors all inspected properties and uses their sales as signals to update her posterior belief over θ in real time. The appendix establishes that the buyer's problem admits a unique bounded value function, that the value function is weakly decreasing in the perceived expiration hazard, and that the optimal stopping set takes a threshold form whose cutoff contracts as beliefs shift toward tighter market states. It further shows that recall purchases occur (almost surely) only at sale-shock times, and that each sale shock both lowers the stopping threshold and weakly raises the hazard of current-house purchase conditional on continuation. All of the main empirical patterns arise from this single Bayesian signal mechanism.

3 Background and Data

3.1 Background

China's real estate market entered a prolonged boom in the early 2000s, a period that lasted until the market slowdown after 2021. The vibrancy of the market, together with the maturation of the real estate brokerage industry, has led a large number of households to participate in the market. In China's housing market, transactions are predominantly intermediated by real estate agents. Both buyers

and sellers are typically represented by separate agents. Listing agents are responsible for posting, maintaining, and updating property information, while buyer agents primarily facilitate property visits and assist buyers during the search process. In practice, these roles often overlap.

Potential buyers can search for suitable properties online and obtain basic information about the listings, such as construction year, location, size and asking price. This process defines the buyer's choice set of properties for in-person visits, constituting the first stage of the housing search process. However, they can only visit properties in person through a real estate agent and must also complete the transaction process via the agent. An in-person inspection is also one of the most critical steps in the home search and purchase process: many unobservable property characteristics can only be assessed on site, and it also provides an opportunity for buyers to negotiate with the seller.

Buyers typically engage in multiple property visits before making a purchase. As a result, housing search is both time-consuming and costly, and purchase decisions may involve previously visited homes rather than only the most recently inspected one. At the same time, the market is dynamic: properties may be purchased by others at any time, while new listings continuously arrive. Buyers therefore face a dynamic search and purchase decision problem. This environment provides an ideal setting to examine how buyer behavior responds to an unexpected market shock—namely, when a previously visited property is purchased by someone else, causing the buyer to lose an option from the choice set.

3.2 Housing Dataset

The housing search and transaction data are obtained from the largest multiple listing service (MLS) platform in China. The platform was established in Beijing and accounts for more than 50% of the housing brokerage market in Beijing, implying broad market coverage. The dataset records all in-person property inspections and completed transactions between August 2015 and July 2016. It further contains detailed housing characteristics, listing price revision histories, and buyer characteristics. The housing characteristics include the property's floor area, numbers of bedrooms, bathrooms, and living rooms, year of construction, unit floor level, building height, apartment complex, and the neighborhood in which the property is located. The transaction dataset records the transaction price, transaction date, and both the seller and buyer associated with each transaction. Notably, the dataset records unique buyer-side information, including buyers' self-reported search preferences at the time of platform registration, such as preferred ranges of prices, floor area, and number of rooms.

3.3 Sample Construction

The paper empirically examines how buyers' search behavior responds when a property in their consideration (choice) set is sold. To do so, we construct a panel dataset at the searcher–day level. Because buyers do not inspect properties every day, the raw records are temporally sparse and omit days without visits. These omitted days are nevertheless informative, as they indicate that the buyer neither conducted inspections nor completed a purchase. To avoid sample selection arising from buyers' endogenous timing of visits, we construct a complete daily panel for each searcher, spanning from the first inspection day to the terminal day. For searchers who complete a purchase, the terminal day corresponds to the transaction day; for those who do not, it corresponds to the last visit day. We treat days with no inspections as periods of zero search activity rather than missing observations.

When examining whether buyers' search intensity and purchase outcomes are affected, we code both the number of inspections and the purchase indicator as zero in days without search activity. By contrast, when analyzing the characteristics of the properties visited, we rely on the original (unexpanded) panel and restrict the sample to days with at least one inspection, since these variables are only defined conditional on a visit.

3.4 Summary Statistics

In our data covering 2015 to 2016, the Beijing housing market exhibits a pronounced boom, with steadily increasing prices and high levels of both transactions and listings. Panels (a) and (b) of Figure 3 indicate that total transactions and visits increase steadily leading up to the Chinese New Year, and then decline gradually due to seasonal factors, though they remain at relatively high levels. In contrast, Panel (c) shows that the total number of new listings remains relatively stable, except for a brief increase following the Chinese New Year. These patterns imply that market tightness has been rising over time, if tightness is defined as the ratio of transactions or visits to new listings or the existing stock of listings. In addition, Panel (d) provides clear evidence of price appreciation, with transaction prices per square meter rising by approximately 30% within a single year. In a tightening market with rapidly rising prices, both buyers' and sellers' beliefs tend to align with prevailing market dynamics. Sellers may gain greater bargaining power and become less inclined to close transactions quickly, while buyers lose bargaining power and are more likely to accelerate their purchase decisions to avoid further price increases.

Table 1 presents the summary statistics of the main variables used in the empirical analysis. Panel (a) reports summary statistics for the buyer–day panel con-

structed based on the procedure described in Section 3.3. In the constructed panel, the average daily purchase probability is 0.663%, and buyers visit about 0.19 property per day on average. Given that buyers do not visit properties every day, this level of search intensity is reasonable. The indicators Above price, Above size, and Above rooms equal one if the price, size, or number of rooms of a visited property exceeds the buyer's stated preference range. As these variables are defined only for days with observed visits, their sample size is approximately 1/10 that of the expanded panel. Panel (b) presents the summary statistics at the buyer level. The data indicate that option loss is common: approximately 30% of buyers experience one option loss. In addition, only about 18% of buyers complete a purchase before the end of our observation period (July 2016). With a daily purchase probability of approximately 0.663%, the implied average search duration is about 27 days, during which buyers make about 5 visits on average.[3]

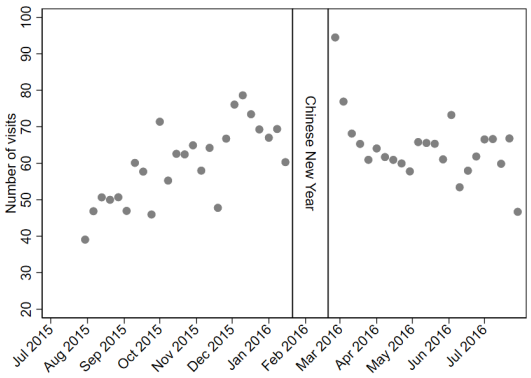
4 Empirical Strategy

4.1 Research Design

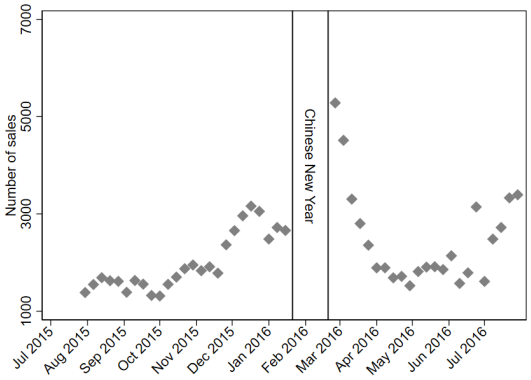
We study how buyers respond when a property they have previously visited is purchased by another buyer. Using the complete history of in-person property visits, we construct for each buyer the set of previously visited properties that remain available at each point in time. We define an *option-loss* event as the sale of one of these previously visited properties while the focal buyer is still active in the market. The buyer is treated from the date of the option-loss event onward. This construction raises two natural questions: whether the buyer is aware that the property has been sold, and why previously visited properties constitute a meaningful set of options. We discuss each in turn.

Why previously visited properties matter. Previously visited properties constitute only a subset of the buyer's broader consideration set, which may also include listings viewed online but never inspected in person. Nevertheless, they represent a particularly salient subset for the purchase decision. An in-person visit requires the buyer to incur time and search costs and allows her to learn important property characteristics that are difficult to assess online, while also providing an opportunity to interact with the seller and evaluate the feasibility of a transaction. Visited properties have therefore already passed an initial screening stage and are options about which the buyer has acquired substantially more information. Moreover, this subset is actively monitored through the buyer–agent relationship. Focusing

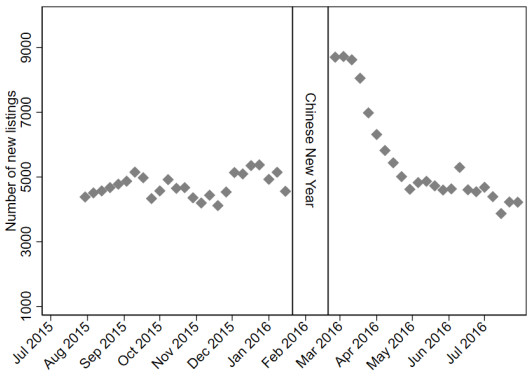
[3]These figures are lower than those documented in other countries. For example, Genesove and Han [2012] document that buyers on average search for 8.2 weeks and visit 10 properties in the US.



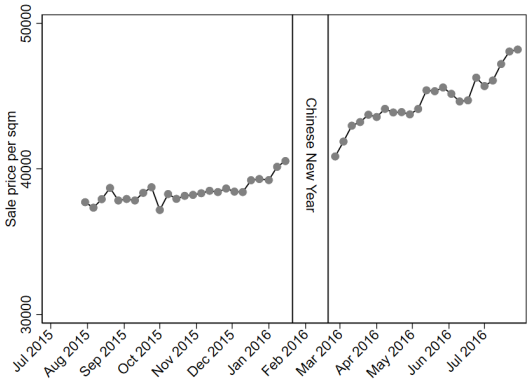
(a) Number of Visits



(b) Number of Sales



(c) Number of New Listings



(d) Sale Prices

Figure 3: The Dynamics of Beijing Housing Market

Notes: This figure plots aggregate trends in Beijing's housing market during the sample period. Panel (a) shows the total number of property visits, Panel (b) shows the total number of sales, Panel (c) shows the total number of new listings, and Panel (d) shows the average transaction price per square meter. Visits and transactions increase before the Chinese New Year and decline afterward, reflecting seasonal patterns, while new listings remain relatively stable except for a temporary increase after the holiday. The upward trend in transaction prices indicates substantial price appreciation during the sample period.

Table 1: Summary Statistics

Variable	Mean	SD	Min	Max	*N*
Panel A: Variables at buyer-day level					
Purchase probability	0.663	8.114	0	100	4,008,432
Number of visits	0.186	0.658	0	10	4,008,432
Recall visits	0.020	0.168	0	8	4,008,432
Non-recall visits	0.165	0.621	0	10	4,008,432
Option Loss	0.331	0.471	0	1	4,008,432
Above price	0.354	0.438	0	1	420,112
Above rooms	0.006	0.070	0	1	420,112
Above size	0.273	0.409	0	1	420,112
Panel B: Buyer characteristics					
Option Loss	0.308	0.462	0	1	151,493
Purchase probability	0.183	0.386	0	1	151,493
Mortgage	0.824	0.381	0	1	151,493
Male	0.510	0.500	0	1	151,493
Search duration	26.98	35.69	2	226	151,493
Total visits	5.016	2.974	1	75	151,493
Preferred rooms (upper)	2.461	0.798	1	6	151,493
Preferred rooms (lower)	2.003	0.703	1	6	151,493
Preferred price (upper)	439.3	245.7	100	1600	151,493
Preferred price (lower)	351.6	189.0	50	1100	151,493
Preferred size (upper)	104.4	40.43	42	280	151,493
Preferred size (lower)	81.15	32.09	30	200	151,493

Notes: This table reports summary statistics for the main variables at buyer-day level and buyer-level. *Purchase probability* indicates whether a searcher makes a purchase in a given day. # *of Visits* is the total number of visits in a day, decomposed into recall and non-recall visits. *Option Loss* indicates whether a previously visited property has been purchased by another buyer. *Above price*, *Above rooms*, and *Above size* indicate whether the visited property exceeds the searcher's stated preference range. Panel B reports additional buyer characteristics. *Option Loss* indicates the share of buyers who experience option loss during the search process. *Mortgage* equals to 1 if the buyer is a mortgage buyer. *Male* equals to 1 if the buyer is a male. *Total visits* indicates the number of visits the buyer made during her search process. *Search duration* measures the time between the first visit and the last visit or purchase. Preferred price is measured in million *CNY*, and preferred size is measured in m^2.

on previously visited properties therefore provides an observable and economically meaningful subset of the buyer's broader consideration set for identifying option-loss events.

How buyers learn about option loss. In-person visits are typically arranged through a buyer agent, who remains involved throughout the search and transaction process. Importantly, agents continue to monitor properties that their clients have visited. When a previously visited property is sold to another buyer, the focal buyer is therefore likely to learn about the sale through her agent, often shortly after the transaction is recorded in the brokerage system. Agents also have strong incentives to facilitate transactions and may use the sale of a previously visited property to encourage buyers to accelerate their search or purchase decisions. Buyers may also receive information through online subscriptions, although we do not observe which properties they follow online. The sale of a previously visited property is therefore an event that is likely to be communicated to and recognized by the buyer.

Figure 4 illustrates the construction of treatment and control observations. There are three types of buyers, denoted by 1, 2, and 3, each of whom visits a set of properties, denoted by A–F. Buyer 1 remains in the market when a previously visited property, A, is purchased by another buyer and exits only afterward, either with or without making a purchase. Buyer 1 is therefore treated from the sale of property A onward. Buyer 2 exits the market before a previously visited property, F, is purchased by another buyer and therefore contributes only untreated observations. Buyer 3 visits three properties, none of which is sold while the buyer remains active in the market, and therefore serves as a control buyer. The gray line segments represent untreated periods, while the thick red segments represent treated periods. Accordingly, our empirical design compares buyer outcomes following option loss with outcomes during untreated search spells.

We first provide descriptive evidence by comparing survival probabilities over the same time-on-market spells between treated and control regions. Figure A1 plots Kaplan–Meier survival estimates for the two groups. The survival probability declines more rapidly over time-on-market in the treated region, indicating that buyers exposed to option loss tend to purchase earlier. The gap between the two curves emerges early and persists over time, providing descriptive evidence that option loss is associated with an acceleration in the timing of purchase.

4.2 Endogeneity and Identification

Exposure to option loss is not randomly assigned: more attractive or competitively priced properties may be more likely to sell, and searchers who inspect such

properties may differ from those who do not experience option loss. Table A1 reports balance tests for buyer characteristics between the treatment and control groups. The results show statistically significant differences in several characteristics. However, except for search duration, the magnitudes of these differences are small, suggesting that treated and control buyers are broadly comparable along observable characteristics. To address the substantial difference in search duration between treated and control buyers, our empirical specifications control for search-duration fixed effects, which we discuss in more detail below.

Our identification strategy follows the logic of nonrandom exposure to exogenous shocks proposed by Borusyak and Hull [2023] by leveraging variations from the timing of the option loss, which is arguably random for the treated buyer. Conditional on a searcher having inspected a property that is at risk of sale, the exact timing of the sale to another buyer is difficult for the searcher to predict. Sellers often negotiate with multiple potential buyers simultaneously, and the transaction date reflects bargaining frictions and agreement timing between the seller and the eventual buyer. Thus, while the likelihood that a property eventually sells may be endogenous, the daily timing of its sale is plausibly exogenous from the perspective of other searchers who previously inspected the property. This timing variation determines treatment status in our setting. A searcher may have inspected a property that is later sold to another buyer; if the sale occurs only after the searcher has exited the market by purchasing another property, the searcher does not experience option loss and remains in the control group. Had the same sale occurred earlier, before the searcher exited the market, the searcher would have been treated. In this sense, quasi-random variation in sale timing assigns otherwise comparable searcher-days to treatment and control status.

We therefore exploit the staggered timing of sales of previously inspected properties as option-loss shocks. Our difference-in-differences design compares the evolution of search and purchase outcomes before and after option loss for affected searchers relative to searchers who have not yet experienced option loss. The identifying assumption is not that affected and unaffected searchers are identical, nor that exposure to properties that may eventually sell is random. Rather, it is that, conditional on the controls and fixed effects included in the specification, the exact timing of option loss is as good as random with respect to potential short-run changes in the searcher's subsequent behavior.

4.3 Model Specification

Diff-in-Diff specification. We define the treatment group as searchers who experience option loss during their search process, while controls are searchers who never experience option loss or do not experience option loss before exiting the

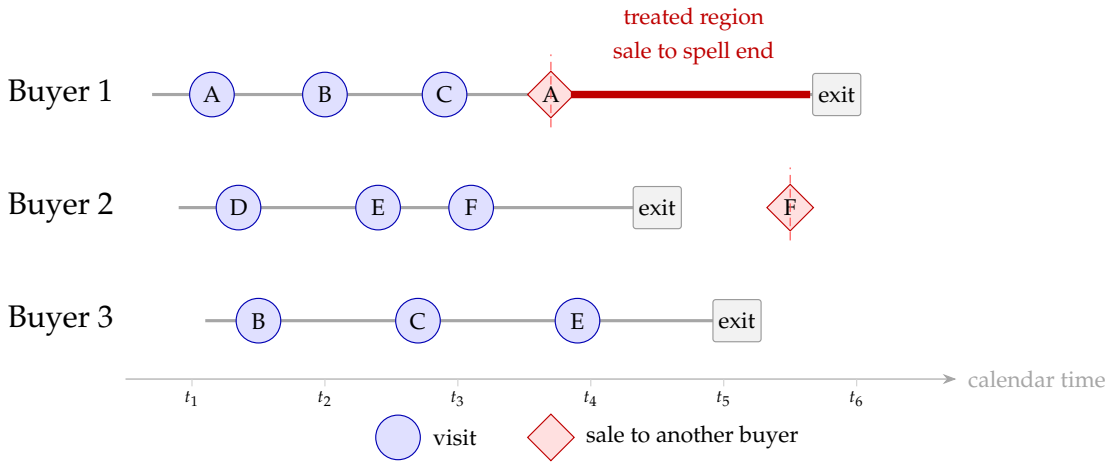


Figure 4: Research Design

Notes: There are three types of buyers, denoted by 1, 2, and 3, each of whom visits a set of properties, denoted by A–F. Buyer 1 remains in the market when a previously visited property, A, is purchased by another buyer and exits only afterward, either with or without making a purchase. Buyer 1 is therefore classified as treated. Buyer 2 exits the market before a previously visited property, F, is purchased by another buyer and contributes only to the pre-treatment period of the treated group. Buyer 3 visits three properties, none of which is sold during the sample period, and therefore serves as a control buyer. The gray line segments represent untreated periods, while the thick red segments represent treated periods.

market. Because a searcher may experience option loss repeatedly within the consideration set, treatment exposure is potentially multiple. To maintain a well-defined treatment timing and comply with the standard DID framework that focuses on a single treatment event, we restrict the sample to buyers who experience option loss only once during the search process. This group accounts for more than 60% of all buyers who ever experience option loss. Including buyers with multiple option loss events may introduce bias, as these buyers are likely to be systematically different, for example by having substantially longer search durations. We therefore define treatment timing as the occurrence of the sale of a previously inspected property. In other words, a buyer is considered treated after the first sale of a previously visited property and is expected to adjust subsequent search behavior thereafter.

We start our estimation with a canonical two-way fixed effect specification:

$$Y_{it} = \alpha + \beta \cdot Option_Loss_{it} + \gamma_i + \eta_t + \delta_{\tau_{it}} + X_i' \eta_t + \epsilon_{it}, \tag{8}$$

where Y_{it} indicates the outcome variables concerning about searcher i's search behavior at day t. The primary focus of the paper is the transaction probability of the searcher. Y_{it} equals 100 if searcher i purchases a property on day t, and 0 otherwise. In addition, we replace the dependent variable with the number of visits by the searcher in order to examine the impact of option loss on visiting behavior. We decompose search behavior into recall and non-recall components, including

recall visits, and non-recall visits. A visit is classified as a recall visit if the buyer revisits a previously inspected property, and as a non-recall visit otherwise. Finally, to examine whether buyers shift toward different types of properties after experiencing option loss, particularly those that deviate from their initial preferences, we construct three dummy variables indicating whether a property's price, size, and number of rooms exceed the buyer's preferred ranges.

$Option_Loss_{it}$ is an indicator equal to one if searcher i has experienced at least one option loss prior to day t, where an option loss occurs when a previously visited property is purchased by another buyer and thus removed from the searcher's choice set. γ_i denotes the searcher fixed effects. η_t is the day fixed effect. $\delta_{\tau_{it}}$ denotes time-on-market fixed effects, with τ_{it} measured as the number of days since buyer i's first property visit. This is crucial in our setting. Because treated and control buyers differ substantially in search duration, their purchase probabilities may differ systematically simply because they are observed at different stages of the search process. We therefore include time-on-market fixed effects, measured by the number of days since the buyer's first property visit. This fixed effect ensures that treated and control buyers are compared at the same search duration, so that the estimated effect is not driven by differences in how long they have been searching.

To account for potential selection of buyers into treatment based on observable characteristics, we flexibly control for differential time trends across buyer types, denoted by $X_i'\eta_t$. These characteristics X_i include gender, payment method (mortgage or cash), preferred neighborhood,[4] and quintiles defined by the searcher's stated price range. Identification therefore comes from within-type comparisons of treated and untreated searchers on the same day, rather than from differences in aggregate market dynamics. To address potential bias arising from negative weighting in staggered difference-in-differences designs, we employ alternative estimators, such as Sun and Abraham [2021], to re-estimate the treatment effects as a robustness check.

Event study and dynamic treatment effects specification. The key identifying assumption in the DID framework is that, absent treatment, outcomes for treated and control groups would follow parallel trends. While this assumption is not directly testable, it is commonly assessed by examining pre-treatment dynamics. In our setting, however, a conventional parallel trends test is not informative. By construction, buyers who eventually experience option loss cannot complete a pur-

[4]Searchers do not report their preferred neighborhoods. However, over 95% of searchers inspect properties in only one neighborhood, which we define as their preferred neighborhood. For searchers who visit multiple neighborhoods, we define the first neighborhood visited as the preferred neighborhood.

chase prior to treatment, implying that their pre-treatment purchase probability is mechanically zero. As a result, pre-treatment trends cannot be meaningfully compared between treated and control groups.

To address this limitation, we exploit variation in the timing of option loss relative to buyers' exit from the market. Specifically, we use buyers whose option loss occurs after they exit the market as a proxy for pre-treatment observations[5]. These buyers are comparable to the treated group in that they are also exposed to the risk of losing a previously viewed property, with the only difference being whether the loss occurs before or after the buyer exits the market. When option loss occurs early, buyers remain active and contribute to post-treatment outcomes; when it occurs late, or buyers exit early, the loss takes place after exit and these observations serve as a proxy for pre-treatment behavior.

Under the assumption that the timing of option loss is as-good-as random conditional on observables, this variation provides a quasi-experimental source of identification. An alternative way to interpret this approach is that we compare the average purchase probability of all buyers who have previously viewed a given property, before and after the property is sold to another buyer, leveraging variation in the timing of the sale.

We estimate the following event study regression for checking the parallel pre-trend assumption and dynamic treatment effects.

$$Y_{it} = \alpha + \sum_{\tau=-30, \tau\neq -1}^{\tau=30} \beta_\tau \mathbb{I}(t - option_loss_day_i = \tau) + \gamma_i + \eta_t + \delta_{\tau_{it}} + \epsilon_{it}, \quad (9)$$

where $option_loss_day_i$ denotes the timing of the option loss event associated with searcher i. This event may occur either before or after the buyer's purchase (or exit) from the market, depending on the relative timing of the loss and the buyer's decision. The remaining variables are defined as in Equation (8).

5 The Main Results

5.1 The Effects of Option Loss on Purchase Probability

The primary focus of this paper is to examine whether buyers' purchase probabilities are affected by option loss. We begin by examining the dynamic effects of option loss and the validity of the parallel-trends assumption. Figure 5 plots the event-study coefficients for purchase probability around the option-loss event.

[5]When estimating other outcomes, such as the number of visits, we continue to use the standard event-study specification. Unlike purchase, these outcomes are not mechanically zero before treatment for treated buyers. We therefore use treated buyers' pre-treatment observations to estimate pre-trends and assess the validity of the parallel trends assumption.

Prior to option loss, there is no evidence of systematic differences in the evolution of purchase probabilities between treated and control buyers. Following option loss, however, treated buyers exhibit an immediate increase in purchase probability relative to the control group, and the effect remains relatively stable over subsequent periods.

The absence of differential pre-trends is important for the causal interpretation of our estimates. One potential concern is that both option loss and buyers' purchase decisions may be jointly driven by changes in local market conditions. For example, a market-wide increase in demand could simultaneously raise the probability that previously visited properties are sold and increase the focal buyer's probability of purchasing. If such common market trends were driving the results, we would expect buyers who subsequently experience option loss to already exhibit relatively higher purchase probabilities before the event. Instead, the increase in purchase probability appears only after option loss, with no corresponding differential trend beforehand. This pattern is therefore inconsistent with a simple common-market-trend explanation and supports the interpretation that the change in purchase behavior is associated with the experience of option loss.

We next estimate the average effect of option loss using equation (8). Table 2 reports the estimated effects on buyers' purchase probability. Column (1) presents a naive specification and shows that buyers experiencing option loss have an approximately 1 percentage point higher purchase probability. This estimate, however, may be confounded by differences in buyers' positions in the search process. Purchase probabilities exhibit substantial duration dependence: buyers at different stages of their search may have systematically different propensities to purchase. Because buyers who experience option loss may have remained in the market longer, comparing them directly with all untreated buyers may therefore confound treatment effects with differences in time on market. To address this concern, column (2) adds day-on-market fixed effects. The identifying comparison is therefore between buyers observed on the same calendar day and at the same stage of their search spell. The estimated treatment effect declines to approximately 0.65 percentage points. Building on this specification, column (3) additionally controls for the number of visits on a given day, while column (4) includes interactions between buyer characteristics and time fixed effects. The estimated coefficients remain highly stable across these specifications, at around 0.6 percentage points.

The magnitude of the estimated effect is economically substantial. The mean purchase probability among untreated observations is approximately 0.89 percentage points.[6] An estimated increase of around 0.6 percentage points therefore cor-

[6]Because treated buyers must remain unpurchased before experiencing option loss, their pre-treatment purchase probability is mechanically zero. We therefore compare the estimated ATT with the mean purchase probability among observations that have not yet experienced treatment.

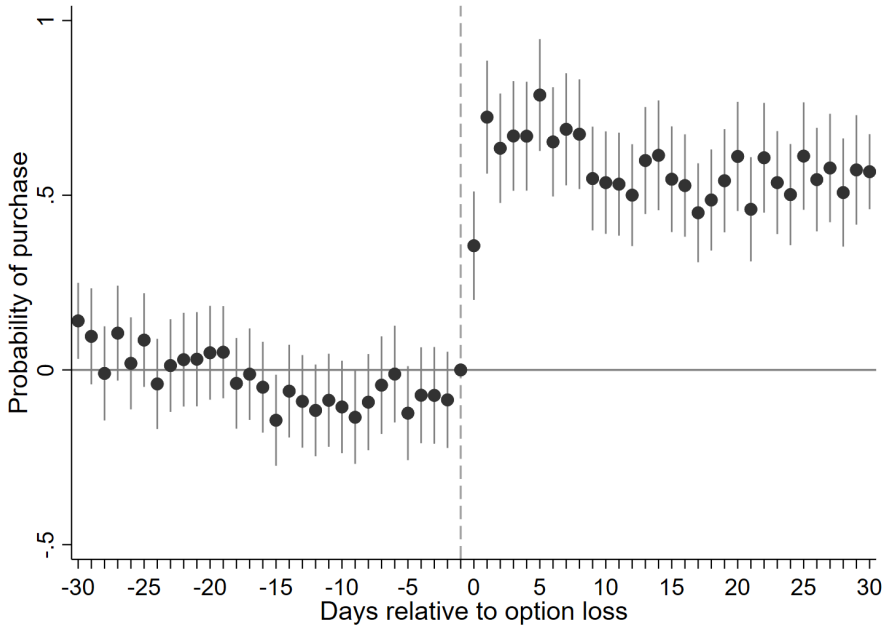


Figure 5: The Dynamic Treatment Effects on Purchase Probability

Notes: This figure reports the event-study coefficients estimated from Equation (9). Vertical bars represent 95 percent confidence intervals. The specification corresponds to Column (2) of Table 2.

responds to roughly a 67% increase relative to the untreated baseline purchase probability.

Our empirical findings are consistent with the predictions of the theoretical model in Section 2. In a sequential search framework with option loss, the possibility of losing previously visited properties reduces the continuation value of waiting and lowers buyers' reservation thresholds, leading to faster purchases and higher purchase probabilities. Moreover, option loss may serve as an informative signal of market tightness, inducing buyers to update their beliefs about future housing availability and market conditions. As a result, buyers become more likely to stop searching and make immediate recall purchases, particularly in a booming market with persistently rising prices.

5.2 The Effects of Option Loss on Other Outcomes

Beyond purchase probabilities, other aspects of buyers' search behavior may also be affected by option loss. This section aims to further discuss how buyers' search behavior can be affected after losing options in the choice set.

These observations form the relevant comparison group because they remain at risk of treatment: had a previously inspected property been sold before the buyer exited the market, they could have become treated.

Table 2: Effects of Option Loss on Purchase Probability

	Purchase = 100			
	(1)	(2)	(3)	(4)
Option Loss	1.045***	0.641***	0.645***	0.595***
	(0.019)	(0.019)	(0.019)	(0.019)
# visits			0.101***	0.100***
			(0.014)	(0.014)
N	4,008,432	4,008,432	4,008,432	4,007,141
R^2	0.150	0.156	0.156	0.180
Searcher F.E.	✓	✓	✓	✓
Day F.E.	✓	✓	✓	✓
Day-on-market F.E.		✓	✓	✓
Payment × Day F.E.				✓
Gender × Day F.E.				✓
Neighbor × Day F.E.				✓
Group × Day F.E.				✓

Notes: Standard errors are clustered at the searcher level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. The dependent variable is the purchase probability at day level. The key explanatory variable, *Option Loss*, indicates that a property previously visited by the searcher has been purchased by another buyer, resulting in the loss of an option from the searcher's choice set. Columns (2)–(4) additionally control for Day-on-market fixed effects, where day-on-market is defined as the number of days elapsed since the searcher's first visit. Columns (3)–(4) additionally control for the number of visits at day level. Column (4) includes interactions between day fixed effects and searcher characteristics, including payment method, gender, neighborhood preference, and preferred price group. All specifications include searcher and day fixed effects.

5.2.1 Visits

This section presents the results on searchers' property inspections. We use the same difference-in-differences specification to examine whether option loss affects buyers' number of visits. The estimation results are reported in Table 3. Columns (1)–(2) show that option loss significantly reduces the total number of visits, with an estimated decline of about 0.04 visits per day. Decomposing visits into recall and non-recall components in Columns (3)–(6) reveals important heterogeneity. The results show that, after option loss, recall visits decline by only 0.002 visits per day on average, whereas non-recall visits decline by about 0.039 visits per day. Relative to their respective sample means, recall visits decline by approximately $(0.002/0.020 \approx 10\%)$, while non-recall visits decline by approximately

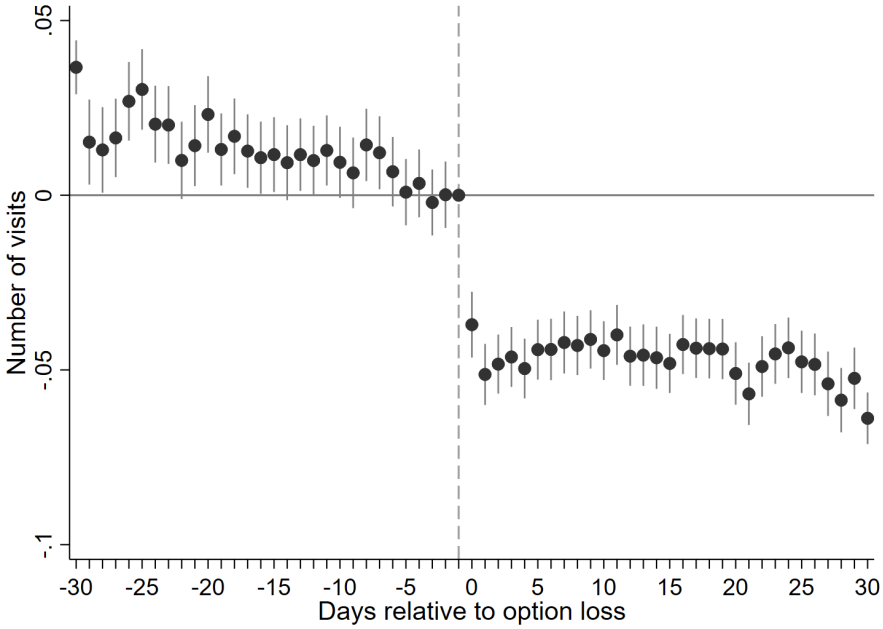


Figure 6: The Dynamic Treatment Effects on Buyer Visits

Notes: This figure reports the event-study coefficients estimated from Equation (9). Vertical bars represent 95 percent confidence intervals. The specification corresponds to Column (1) of Table 3.

$(0.038/0.165 \approx 23\%)$. Thus, the proportional decline in non-recall visits is more than twice as large as the decline in recall visits. These results suggest that option loss primarily reduces buyers' exploration of new properties, while visits to previously inspected properties are much less affected.

We also present the dynamic treatment effects plots in Figure 6 to interpret our estimates as causal effects. They show that the treatment and control groups are not perfectly parallel, with a slight downward trend in the pre-treatment period. However, pre-treatment trends are relatively stable overall, and the post-treatment effects remain clearly visible and statistically significant.

The explanation is that option loss conveys a signal of market tightness, prompting buyers to update their expectations and accelerate their purchase decisions, as reflected in the results above. In the continuous-time model, Proposition D.10 formalizes this implication: for buyers whose recalled option survives and whose state is shifted into the stopping region by the sale shock, the remaining number of future inspections falls to zero. The model therefore predicts a reduction in future search, especially in exploration of new properties. This prediction matches the empirical pattern that non-recall visits decline much more than recall visits. Relative to new-property search, visits to previously inspected properties are much less affected, consistent with buyers shifting away from exploration and toward finalizing decisions within their existing choice set.

Table 3: Effects of Option Loss on Buyer Visits

	Visits		Recall Visits		Non-recall Visits	
	(1)	(2)	(3)	(4)	(5)	(6)
Option Loss	-0.038***	-0.039***	-0.002***	-0.002***	-0.036***	-0.038***
	(0.001)	(0.001)	(0.000)	(0.000)	(0.001)	(0.001)
N	4,008,432	4,007,141	4,008,432	4,007,141	4,008,432	4,007,141
R^2	0.443	0.456	0.170	0.188	0.466	0.478
Searcher F.E.	✓	✓	✓	✓	✓	✓
Day F.E.	✓	✓	✓	✓	✓	✓
Day-on-market F.E.	✓	✓	✓	✓	✓	✓
Payment × Day F.E.		✓		✓		✓
Gender × Day F.E.		✓		✓		✓
Neighbor × Day F.E.		✓		✓		✓
Group × Day F.E.		✓		✓		✓

Notes: Standard errors are clustered at the searcher level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. The dependent variables are the number of visits at the day level, decomposed into total visits, visits to previously viewed (recalled) properties, and visits to new (non-recalled) properties. The key explanatory variable, *Option Loss*, indicates that a property previously visited by the searcher has been purchased by another buyer, resulting in the loss of an option from the searcher's choice set. Odd-numbered columns report specifications with searcher, day, and day-on-market fixed effects. Even-numbered columns additionally include interactions between day fixed effects and searcher characteristics, including payment method, gender, neighborhood preference, and preferred price group.

5.2.2 Visited Houses

Given that option loss increases buyers' purchase probabilities, it is natural to ask whether buyers adjust their search behavior along another margin—namely, whether they become more willing to consider properties outside their initially stated preference ranges. If buyers seek to complete a transaction more quickly after experiencing option loss, they may relax their search criteria and visit properties that do not strictly match their original preferences. To examine this, we construct indicators for whether the visited property's price, size, or number of rooms exceeds the buyer's registered preference range.

The results in Table 4 show that option loss significantly increases the likelihood that buyers visit properties above their preferred price and size ranges. The estimated coefficients are around 1.2–1.7 percentage points, indicating economically meaningful shifts in search behavior. By contrast, the effect on the number of rooms is significantly negative and quantitatively small. The parallel trends plot in Figure 7 shows that the treatment and control groups are comparable prior to

Table 4: Effects on Visiting Houses Above Preferred Range

	Above Price		Above Size		Above Rooms	
	(1)	(2)	(3)	(4)	(5)	(6)
Option Loss	0.017***	0.016***	0.014***	0.012***	-0.001**	-0.001**
	(0.003)	(0.003)	(0.003)	(0.003)	(0.000)	(0.000)
N	413,065	395,937	413,065	395,937	413,065	395,937
R^2	0.585	0.676	0.577	0.669	0.607	0.687
Searcher F.E.	✓	✓	✓	✓	✓	✓
Day F.E.	✓	✓	✓	✓	✓	✓
Day-on-market F.E.	✓	✓	✓	✓	✓	✓
Payment × Day		✓		✓		✓
Gender × Day		✓		✓		✓
Neighbor × Day		✓		✓		✓
Group × Day		✓		✓		✓

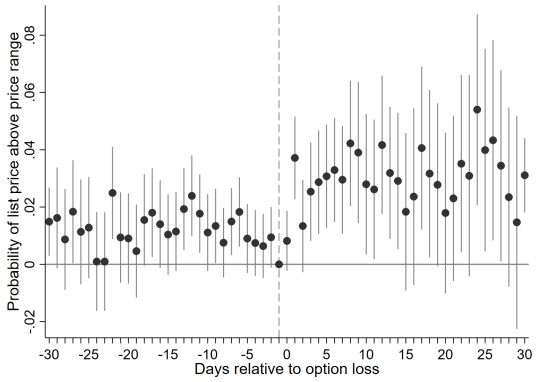
Notes: Standard errors are clustered at the searcher level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. The dependent variables are indicators equal to one if the property viewed in a given day exceeds the buyer's stated preference in price, size, or number of rooms, respectively. The key explanatory variable, *Option Loss*, indicates that a property previously visited by the searcher has been purchased by another buyer, resulting in the loss of an option from the searcher's choice set. Odd-numbered columns report specifications with searcher, day, and day-on-market fixed effects. Even-numbered columns additionally include interactions between day fixed effects and searcher characteristics, including payment method, gender, neighborhood preference, and preferred price group.

the occurrence of option loss. Following option loss, a clear divergence emerges immediately. This pattern is consistent with a causal effect of option loss on the three indicators capturing whether choices exceed buyers' stated ranges.

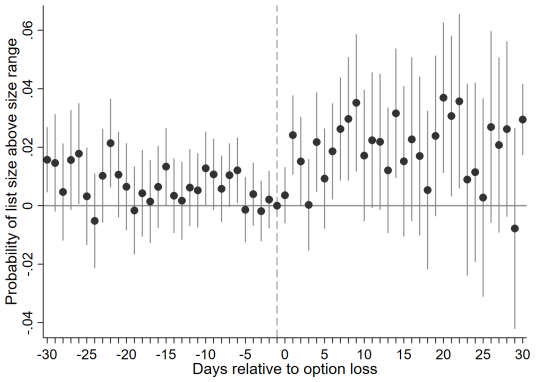
These findings suggest that, following option loss, buyers adjust their search strategy by relaxing certain dimensions of their preferences, particularly along price and size. One interpretation is that option loss signals a tightening market, leading buyers to update their expectations and prioritize transaction completion over strict adherence to initial preferences. As a result, buyers expand their search to include higher-priced or larger properties that may offer a higher likelihood of successful matching.

5.2.3 Sale Prices

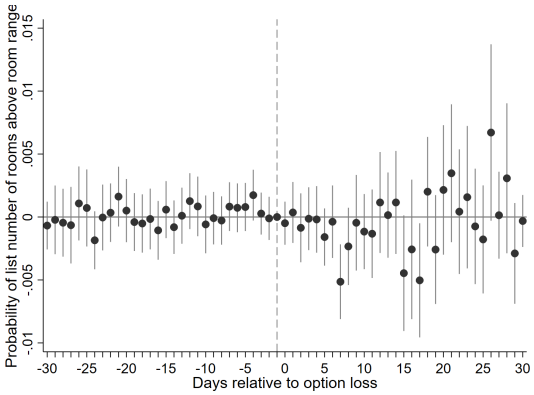
We examine a range of outcomes associated with a buyer's loss of an option. However, transaction price is not included because it is observed only at the time of purchase and therefore cannot be identified within our DiD framework. We directly



(a) Above preferred price range



(b) Above preferred size range



(c) Above preferred number of rooms

Figure 7: The dynamic treatment effects on visited house characteristics

Notes: This figure reports event-study estimates of the effect of option loss on whether buyers visit properties outside their stated preference ranges. The outcomes indicate whether a visited property exceeds the buyer's stated price range, size range, or preferred number of rooms. The estimation sample is restricted to buyer-day observations with at least one recorded visit. Vertical bars represent 95 percent confidence intervals.

regress transaction prices on the treatment indicators at buyer level only with buyers who bought a property in the sample. The results are presented in Table 5. We show different specifications controlling for neighborhood fixed effects, day fixed effects, housing attributes, such as the size, building age, height, floor level, and number of rooms in all the specifications. Column (1) shows that, conditional on the same housing characteristics, buyers who experience option loss purchase homes at prices approximately 0.5% higher than those paid by buyers who do not experience option loss. We progressively control for buyers' preferred price range, the upper bound price and lower bound price, in column (2). We examine whether buyers who experience a option loss end up purchasing homes that are more expensive or not relative to their initial price preference range. We find that buyers who experience a option loss purchase homes that are significantly more expensive relative to their initial preferences, with the average transaction price about

1.1% higher.

Whether a buyer experiences an option loss largely depends on how long they searched and how many properties they inspected. Buyers who search more intensively are therefore more likely to encounter a option loss, implying that the treatment variable may partly capture the effect of search intensity or time spent in the market. To isolate this component, we control for the total number of properties the buyer has visited during their search process and the total search duration in column (3). The results show that the effect shrinks to 0.9% and still significant.

Finally, we also control for the seller's initial list price. This specification is intended to test whether buyers who experience a option loss lose bargaining power in negotiations with sellers. If, conditional on the same list price, a buyer who has experienced an option loss pays a higher transaction price than a buyer who has not, this would indicate that the shock reduces the buyer's bargaining power. The results indicate that, conditional on the same list price, buyers who have lost an option pay on average about 0.1% more than those who have not, and the difference is statistically significant. This suggests that buyers who lose an option appear to have weaker bargaining power: when facing properties with the same list price, they end up paying a higher transaction price.

This pattern is consistent with the price implication of the continuous-time model discussed in Appendix E.8.3. Although prices are not modeled as primitives, the model implies that a sale shock raises the perceived hazard, lowers the value of waiting, and reduces the buyer's reservation surplus. If net surplus is written as $u = q - p$, where q is housing quality and p is the transaction price, then accepting a lower surplus for a given quality corresponds to a higher willingness to pay or a weaker bargaining position. The positive association between option loss and transaction prices can therefore be interpreted as the empirical counterpart of the model's lower-reservation-surplus prediction.

In summary, there is robust evidence that higher purchase price is associated with losing inspected houses. At the same time, these findings should be interpreted with caution. We cannot fully address the endogeneity of whether a buyer experiences a option loss, nor can we establish a causal effect. Although we control for a rich set of buyer characteristics, property attributes, and fixed effects, the results should therefore be viewed as correlations and treated as suggestive evidence.

5.3 Within-property Design

One remaining concern with the baseline design is that treated and control buyers need not have visited the same properties. Differences in purchase behavior may therefore partly reflect heterogeneity in the types of properties buyers inspect or

Table 5: Correlation Between Option Loss and Transaction Prices

	Log Transaction Price			
	(1)	(2)	(3)	(4)
Option Loss	0.005**	0.011***	0.009***	0.001***
	(0.002)	(0.002)	(0.002)	(0.000)
Upper Stated Price (log)		0.309***	0.308***	0.004***
		(0.010)	(0.010)	(0.001)
Lower Stated Price (log)		0.130***	0.131***	0.005***
		(0.008)	(0.008)	(0.001)
Gender		-0.004***	-0.004**	0.000
		(0.001)	(0.002)	(0.000)
Mortgage		0.010***	0.009***	0.002***
		(0.003)	(0.003)	(0.000)
# visits (log)			-0.003**	
			(0.002)	
Search Duration (log)			0.009***	
			(0.001)	
List Price (log)				0.989***
				(0.002)
N	47,865	47,865	32,478	47,863
R^2	0.871	0.916	0.912	0.997
House Attributes	✓	✓	✓	✓
Neighbor F.E.	✓	✓	✓	✓
Transaction-week F.E.	✓	✓	✓	✓

Notes: Standard errors are clustered at the neighborhood level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. The dependent variable is the logarithm of the transaction price. The key explanatory variable, *Option Loss*, indicates that a property previously visited by the searcher has been purchased by another buyer. Columns (2)–(3) progressively control for buyers' preferred price range, buyer search intensity, and search duration. Column (4) further controls for the list price of the purchased property. All specifications include buyer's neighborhood fixed effects, transaction-week fixed effects, and housing attributes as controls.

in the local markets in which those properties are located. To address this concern, we implement a complementary within-property design. For each property that is eventually sold and generates an option-loss event, we identify all buyers who had previously visited that property and trace their search behavior over a 30-day window before and after the property's sale date. In this design, buyers associated with the same event have all inspected the same property, so the focal property and its associated location and characteristics are held fixed. We examine two daily outcomes: the number of additional property visits made by the buyer and an indicator for whether the buyer purchases a property on that day.

Formally, let T_j denote the sale date of property j and define event time as $k = t - T_j$. For each property j and event day $k \in [-30, 30]$, we calculate

$$\bar{Y}_{jk} = \frac{1}{N_j} \sum_{i \in \mathcal{V}_j} Y_{i,T_j+k},$$

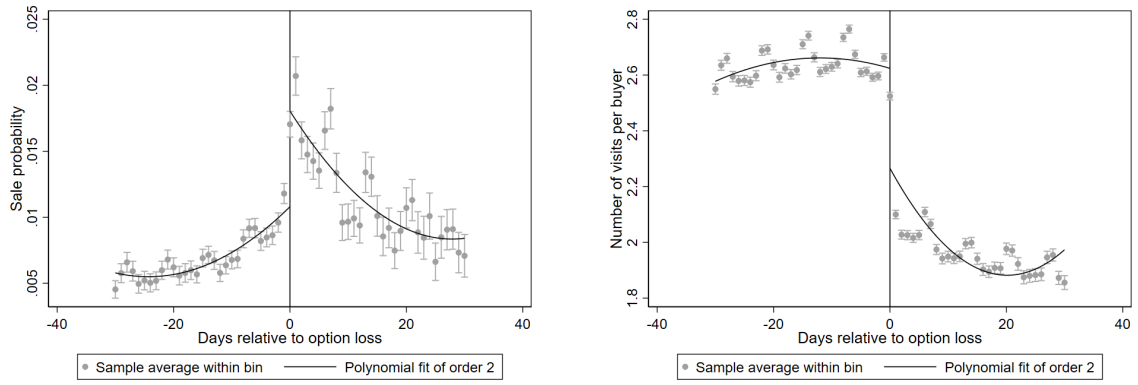
where $\mathcal{V}_j$ denotes the set of buyers who visited property j before its sale, N_j is the number of such buyers, and Y_{it} denotes either the number of property visits made by buyer i on date t or an indicator for whether the buyer purchases a property on that date. We then average $\bar{Y}_{jk}$ across option-loss properties to obtain the mean outcome at each event time:

$$\bar{Y}_k = \frac{1}{J_k} \sum_j \bar{Y}_{jk}.$$

We plot $\bar{Y}_k$ from 30 days before to 30 days after the sale date of the focal property.

The purpose of this exercise is to examine whether the sale of the same previously visited property is followed by a discrete change in the behavior of buyers who had inspected it. In contrast to the baseline design, the comparison no longer relies on buyers having visited potentially different properties, substantially reducing concerns that the estimated response reflects differences in property quality, attractiveness, or local market conditions. If the baseline results are driven by option loss, we should observe little systematic change in buyer behavior before the focal property's sale, followed by an increase in purchase probability and a corresponding adjustment in search activity around or after the sale date.

Figure 8 reports the results from this within-property event-time exercise. Around the sale date of the focal property, both outcomes exhibit a sharp change: the average purchase probability among buyers who had previously visited the property rises markedly, while their average number of visits declines. This pattern is consistent with the interpretation that the sale of a previously inspected property accelerates buyers' purchase decisions and reduces subsequent search activity. We therefore interpret this within-property event-time analysis as complementary evidence that isolates variation associated with the timing of the sale of a common previously inspected property.



(a) Purchase probability

(b) Number of visits

Figure 8: Buyer Behavior around the Sale of a Previously Visited Property

Notes: This figure presents descriptive event-time patterns in buyer behavior around the sale of a previously visited property. For each property that is eventually sold, we identify all buyers who visited the property before its sale and track their search and purchase behavior over a 30-day window before and after the sale date. Panel (a) reports the average daily purchase probability, while Panel (b) reports the average daily number of property visits. Event time is defined relative to the sale date of the focal property, with day 0 indicating the date on which the property is sold. For the visit outcome, we restrict the calculation to buyer-day observations with at least one recorded visit. Including buyer-days with no visits would introduce a large mass of zeros and make the aggregation less informative. As a result, the average number of visits reported in Panel (b) is substantially higher than the unconditional average number of visits reported in the descriptive statistics. The figure reports raw averages and does not include additional controls or fixed effects.

5.4 Ruling Out Local Demand Shocks

A potential threat to identification is that the sale timing of previously inspected properties may be correlated with the searcher's own purchase timing for reasons unrelated to option loss. In particular, both events may be driven by common local demand shocks. For example, a positive demand shock may cause previously inspected properties to sell faster while simultaneously increasing the searcher's likelihood of purchasing. Under this alternative explanation, the observed increase in purchase probability following option loss would reflect common market conditions rather than a causal response to losing an option. We first address this concern using the event-study specification. If common demand shocks or other unobserved factors jointly drive property sales and buyer purchases, treated searchers should exhibit higher purchase probabilities even before option loss occurs. In contrast, our identifying assumption requires treated and control searchers to follow parallel trends prior to option loss, with any increase in purchase probability emerging only after the event. The event-study results are consistent with this pattern: purchase probabilities are similar between treated and control searchers before option loss and increase only after the previously inspected property is sold.

To provide additional evidence against the local-demand-shock explanation,

we conduct a placebo test based on unvisited properties. The key distinction underlying our interpretation is that the property generating the actual treatment had previously been visited by the buyer and had therefore entered the buyer's salient choice set. Its sale removes an option that the buyer had actively considered. By contrast, the sale of an otherwise similar property that was never visited by the buyer should reflect similar local market conditions without generating option loss. If local demand shocks were responsible for the main effect, the sale of such unvisited properties should also be followed by an increase in the focal buyer's purchase probability. Conversely, finding little or no response to these placebo sales would be more consistent with the interpretation that the main effect operates through the loss of a previously considered option.

For each treated buyer, we therefore identify unvisited properties that are comparable to the property generating the actual option-loss event and use their sale dates to construct placebo treatment timings. Specifically, we define a property as comparable if it was listed in the same week and is located in the same apartment complex as the actual option-loss property. In our setting, an apartment complex typically consists of several residential buildings within a gated community, developed by the same developer and constructed around the same time. Housing units within a complex are highly standardized, with differences arising mainly from unit size and the number of bedrooms, while residents share the same location and amenities. Properties within the same complex are therefore close substitutes and are exposed to very similar local market conditions. Because the placebo properties were never visited by the focal buyer, however, their sale does not remove an option from the buyer's visited choice set. This design allows us to compare the effect of losing a previously visited property with the effect of nearby sales that proxy for local demand conditions but do not generate option loss.

The first two columns of Table 6 report the effects of the sale of non-visited but otherwise similar neighboring properties. Because a suitable placebo property cannot be identified for every treated buyer, the placebo construction results in some sample attrition. To facilitate a direct comparison, Columns (3) and (4) re-estimate the baseline specification on this same restricted sample, using the actual option-loss event as the treatment. In Column 1, the coefficient on the placebo treatment falls substantially to 0.17, although it remains statistically significant. After adding a rich set of high-dimensional fixed effects in Column 2, the coefficient declines further to 0.08 and is no longer statistically significant. Figure A2 reports the event-study estimates corresponding to Column (1). The dynamic effects are largely insignificant, providing further evidence against local demand shocks as the main driver of our baseline results.

In contrast, Columns (3) and (4) show that the effect of actual option loss remains large in the restricted sample, ranging from 0.53 to 0.58, and is very sim-

ilar to the baseline estimates obtained from the full sample. The placebo treatment remains statistically significant in some less saturated specifications. One possible explanation is that buyers may also pay attention to other listings in the same building, even if they never visited them in person. The sale of such properties may therefore still convey relevant information about local market conditions or the availability of close substitutes, so the placebo treatment is unlikely to be completely behaviorally irrelevant. Nevertheless, the placebo effect attenuates substantially once we impose richer controls, whereas the effect of actual option loss remains large and stable. This comparison suggests that the attenuation of the placebo effect is not driven by sample attrition. More importantly, the sharp contrast between the placebo and actual option-loss estimates provides evidence against the view that our baseline results are primarily driven by local demand shocks and supports our proposed mechanism: the sale of a property has a substantially stronger effect on buyer behavior when that property had previously been visited and had therefore entered the buyer's choice set.

5.5 Robustness

We conduct a series of complementary analyses in Appendix A to assess the robustness of the baseline findings and address alternative identification concerns. First, we estimate an extended Cox proportional hazard model that explicitly accounts for duration dependence in buyers' purchase hazards and find an effect of option loss that is highly consistent with the baseline DID estimates. Second, we examine potential bias arising from heterogeneous treatment effects under staggered treatment adoption by inspecting the TWFE weighting structure and implementing a stacked DID/event-study design following Sun and Abraham; both exercises yield estimates similar to the baseline results. Third, we re-estimate the model using only buyer-days with at least one recorded visit, which allows us to study purchase probability conditional on visits and confirms that the increase in purchase probability is not driven by the imputation of zero outcomes on non-visit days. Finally, we conduct a permutation test that randomly reassigns treatment status and timing; the resulting placebo estimates are centered around zero, whereas the actual estimate lies far in the right tail of the placebo distribution. Together, these exercises show that the main result is robust to alternative modeling approaches, heterogeneous treatment effects, alternative sample constructions, and random treatment assignment.

Table 6: Effects of the Sale of Non-visited Neighbors

	Purchase			
	Placebo Option Loss		Actual Option Loss	
	(1)	(2)	(3)	(4)
Option Loss	0.173***	0.083	0.579***	0.528***
	(0.051)	(0.056)	(0.043)	(0.046)
# visits		0.147***		0.151***
		(0.024)		(0.024)
N	1,347,370	1,343,137	1,347,370	1,343,137
R^2	0.172	0.234	0.172	0.234
Searcher F.E.	✓	✓	✓	✓
Day F.E.	✓	✓	✓	✓
Day-on-market F.E.	✓	✓	✓	✓
Payment × Day F.E.		✓		✓
Gender × Day F.E.		✓		✓
Neighbor × Day F.E.		✓		✓
Group × Day F.E.		✓		✓

Notes: Standard errors are clustered at the searcher level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. The dependent variable is the purchase probability at day level. The key explanatory variable, *Option Loss*, indicates that a property previously visited by the searcher has been purchased by another buyer, resulting in the loss of an option from the searcher's choice set. Columns (2)–(4) additionally control for Day-on-market fixed effects, where day-on-market is defined as the number of days elapsed since the searcher's first visit. Columns (3)–(4) additionally control for the number of visits at day level. Column (4) includes interactions between day fixed effects and searcher characteristics, including payment method, gender, neighborhood preference, and preferred price group. All specifications include searcher and day fixed effects.

6 Heterogeneity, Mechanism, and Cascades

6.1 Heterogeneity

First, we explore the heterogeneity of buyers in terms of their payment methods, gender, and price segments. The results are presented in Table 7.

First, column (1) shows that mortgage buyers respond more strongly to option loss than cash buyers. Cash buyers enjoy a stronger bargaining position in the housing market, and are typically associated with lower transaction prices and shorter time to sale [Han and Hong, 2024]. As a result, cash buyers typically have access to a larger set of feasible options and are less likely to be constrained by option loss when making their purchase decisions. Their stronger bargaining position and higher acceptance probability allow them to remain selective, even after losing an option, reducing the need to accelerate purchase. In contrast, mortgage buyers face tighter constraints and a smaller effective choice set. The loss of an option therefore represents a more substantial reduction in available opportunities, prompting them to adjust their expectations and expedite their purchasing decisions. Column (2) shows that there is no discrepancy in terms of the gender of the searcher.

Column (3) shows that buyers who have larger preferred price experience smaller increase of purchase probability after the option loss. One possible explanation is that option loss conveys a signal of a tight and active market, prompting buyers to update their expectations and accelerate their purchase decisions. Those targeting higher-priced properties typically face a larger and less constrained choice set, which provides them with more flexibility and reduces the urgency to act. In contrast, buyers searching in lower-priced segments face a more limited set of available options and are therefore more strongly affected by the loss of an option. As a result, they exhibit a larger increase in purchase probability in response to option loss.

Then we explore the heterogeneity of the treated house. First, we examine whether the price-to-quality ratio of the lost property, measured by price per square meter, has differential effects on buyers' purchase probability. The results in column (4) show that the loss of properties with higher prices per square meters has a smaller effect on buyers' purchase probabilities, whereas the loss of properties with lower price per square meters leads to a larger increase in purchase probability. The sale of more attractive properties conveys a stronger signal of market tightness and competition. When such properties are purchased by other buyers, it indicates that high-quality opportunities are being quickly absorbed, prompting buyers to update their expectations and accelerate their purchase decisions. By contrast, the sale of less attractive, high unit-price properties carries weaker infor-

mational content and therefore induces a smaller behavioral response.

We next interact the option loss indicator with a dummy variable indicating whether the lost property's price, floor area, and number of rooms fall within the buyer's preferred ranges. The results are reported in columns (5) of Table 7. We expect that the impact of option loss is stronger when the lost property better matches the buyer's preferences, as such losses are more relevant for the buyer's decision-making and therefore induce a stronger behavioral response. The estimated coefficients are consistent with our expectations. When the lost property falls within the buyer's preferred ranges in terms of price, floor area, or number of rooms, the increase in purchase probability is significantly larger compared to the loss of properties outside the buyer's preferred ranges.

6.2 Mechanism

The baseline results establish that option loss increases purchase probability. We now examine whether the empirical patterns are consistent with the signal-based mechanism embedded in the theoretical framework developed in Section 2. In the model, option loss affects buyers not only by removing one property from their monitored set, but also by providing information about market tightness. The sale of a previously visited house signals that attractive options may be claimed quickly by competing buyers. This signal reduces the value of continued search and increases the incentive to purchase sooner. The mechanism analysis below therefore serves as a further validation of the model: if option loss operates through this signal-based channel, the purchase response should be stronger when the sale conveys more information about market tightness and when buyers are more exposed to the risk of losing available options.

Market tightness. Our model incorporates an additional mechanism whereby the sale of other properties serves as a signal of market tightness, thereby affecting buyers' subsequent search and stopping decisions. If this mechanism is indeed at work, we would expect the effect of option loss to be amplified when the broader market environment corroborates the signal. In a tight market, the sale of a previously visited property reinforces the impression that attractive options are being quickly absorbed, making the signal more diagnostic of high demand. In a slack market, a single sale is less informative about overall conditions and should induce a weaker response. We test this prediction by interacting the option loss indicator with a measure of prevailing market tightness, defined as the neighborhood-level ratio of transactions to new listings around the time of the shock.

The results, reported in columns (1) and (2) of Table 8, confirm this prediction. The interaction between option loss and market tightness is positive and highly

Table 7: Heterogeneous Effects of Buyer and House Characteristics

	Purchase=100				
	(1)	(2)	(3)	(4)	(5)
Option Loss × Mortgage	0.196***				
	(0.041)				
Option Loss × Female		0.018			
		(0.033)			
Option Loss × Log preferred price			-0.767***		
			(0.034)		
Option Loss × Log (price per m^2)				-0.549***	
				(0.058)	
Option Loss × Within_range					0.209***
					(0.050)
Option Loss	0.479***	0.632***	5.173***	6.498***	0.453***
	(0.038)	(0.025)	(0.206)	(0.617)	(0.048)
N	4,008,432	4,008,432	4,008,432	4,008,432	4,008,432
R^2	0.156	0.156	0.156	0.156	0.156
Searcher F.E.	✓	✓	✓	✓	✓
Day F.E.	✓	✓	✓	✓	✓
Day-on-market F.E.	✓	✓	✓	✓	✓

Notes: Standard errors are clustered at the searcher level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. The dependent variable is a purchase indicator equal to one if the searcher purchases a property in a given day. The key explanatory variable, *Option Loss*, indicates that a property previously visited by the searcher has been purchased by another buyer, resulting in the loss of an option from the searcher's choice set. Each column interacts the Option Loss indicator with a different buyer or property characteristic, including mortgage usage, gender, preferred price level, the price per square meter of the lost property, and an indicator for whether the lost property falls within the searcher's stated price, room, or size range. All specifications include searcher, day, and day-on-market fixed effects.

significant. Buyers who experience option loss in a tight market increase their purchase probability substantially more than buyers who experience the same shock in a slack market. This pattern is consistent with the signal interpretation: the informational content of a sale event is amplified when it is consistent with the prevailing market environment.

To verify the pre-trend assumption in the mechanism analysis and ensure that the observed interaction effects are not driven by differential pre-trends between tighter and looser markets, we split treated buyers into 4 quartiles based on the level of market tightness. We then separately present the parallel trends and dynamic treatment effects for each group relative to the control group. The results presented in Figure A7 show that although the parallel trends assumption is not perfectly satisfied, the dynamic treatment effects differ significantly across market tightness groups. This suggests that the interaction with market tightness is unlikely to be fully driven by pre-existing differential trends.

The size of the choice set. When a previously visited property is sold, the buyer is affected through two distinct channels that work in opposite directions. The first is the signal channel: the sale reveals information about market tightness, which lowers the value of continued search and increases purchase probability. This channel operates regardless of which specific property is sold, because the informational content of the event depends on the fact that a sale occurred, not on the identity of the sold property. The second channel works against purchase: if the sold property happened to be the best option in the buyer's recalled set, then the buyer loses her most attractive fallback, which reduces the likelihood of a recall purchase. These two channels have different implications for how the effect of option loss varies with the size of the buyer's choice set. The signal channel does not depend on which property is sold. The best-option-loss channel, however, depends critically on the probability that the sold property is the buyer's best recalled option. When the buyer has visited many properties, the probability that any particular sold property is her best option is small. When she has visited only a few, the probability is large. As the choice set grows, the negative channel therefore becomes weaker relative to the positive signal channel, and the net effect of option loss on purchase probability should increase with the size of the choice set.

A related behavioral interpretation is that option loss may reduce choice overload. When the choice set is large, buyers may find it harder to compare alternatives and may default to deferring the purchase decision. Removing one option could then simplify the menu, lower the cognitive cost of choice, and make purchase more likely. However, this channel implies the opposite comparative static with respect to choice-set size: losing one option should reduce complexity less when the initial choice set is large, because the relative contraction of the menu is

smaller. A pure choice-overload explanation therefore predicts a weaker marginal effect of one sale shock for buyers with larger choice sets. By contrast, our signal-based mechanism predicts amplification by choice-set size. In the continuous-time model, Proposition D.9 shows that, conditional on observing a sale shock, the immediate recall-purchase probability is increasing in the number of monitored houses N, because the probability that the sale does not remove the buyer's best recalled option is $1 - 1/N$. As N rises, the sale shock still conveys information about market tightness, while the probability that it mechanically destroys the buyer's best fallback option falls. The model therefore predicts that the option-loss effect should be stronger for buyers with larger choice sets.

We test this prediction by interacting the option loss indicator with the number of previously visited properties still available to the buyer at the time of the shock. The results, reported in columns (3) and (4) of Table 8, are consistent with this reasoning. The interaction is positive and statistically significant, indicating that buyers with larger choice sets respond more strongly to option loss. This finding is difficult to reconcile with an account in which option loss matters only because it removes an alternative. Under such an account, losing one option out of many should matter less, not more. The positive interaction is, however, a natural prediction of the signal-based mechanism combined with the diminishing probability of losing the best option. For buyers with large choice sets, the sale of any single property is very unlikely to remove their best option, so the negative channel is negligible and the signal channel operates at nearly full force. For buyers with small choice sets, the same sale event has a substantial probability of eliminating their best recalled property, partially offsetting the positive signal effect.

Similarly, we perform an analysis by splitting the sample into 4 groups based on the size of the choice set. Figure A8 shows buyers with larger choice sets experience a larger increase in purchase probability following option loss. Although the parallel trends assumption is not perfectly satisfied, dynamic treatment effects differ significantly across groups. In particular, buyers with relatively small choice sets exhibit substantially larger differences in pre-treatment purchase probability trends relative to the control group. This suggests that the interaction with choice set size is unlikely to be fully driven by differential pre-trends.

6.3 Transaction Cascades

The empirical findings in this section suggest that option loss can affect not only individual purchase decisions but also aggregate market activity. Three findings point to this possibility. First, option loss has a sizable effect on individual purchase probabilities. Second, about 30% of buyers experience option loss during their search. Third, the effect can propagate through transaction cascades: option

Table 8: The Role of Market Tightness and Choice Set Size

	Purchase=100			
	(1)	(2)	(3)	(4)
Option Loss × Tightness	0.016***	0.014***		
	(0.002)	(0.002)		
Option Loss × Choice set size			0.180***	0.190***
			(0.010)	(0.010)
Option Loss	0.258***	0.266***	-0.112***	-0.206***
	(0.036)	(0.038)	(0.039)	(0.039)
N	3,885,606	3,884,182	4,007,777	4,006,486
R^2	0.157	0.181	0.156	0.180
Searcher F.E.	✓	✓	✓	✓
Day F.E.	✓	✓	✓	✓
Day-on-market F.E.	✓	✓	✓	✓
Payment × Day F.E.		✓		✓
Gender × Day F.E.		✓		✓
Neighbor × Day F.E.		✓		✓
Group × Day F.E.		✓		✓

Notes: Standard errors are clustered at the searcher level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. The dependent variable is the purchase probability at the day level. The key explanatory variable, *Option Loss*, indicates that a property previously visited by the searcher has been purchased by another buyer, resulting in the loss of an option from the searcher's choice set. *Tightness* is defined as the number of transactions divided by the number of listings (per 1,000 units) in the neighborhood of the lost property and in the day of the option loss. *Choice set* measures the number of properties the buyer has visited prior to the treatment. Columns (2) and (4) additionally include interactions between day fixed effects and searcher characteristics, including payment method, gender, neighborhood preference, and preferred price group. All specifications include searcher, day, and day-on-market fixed effects.

loss induces affected buyers to purchase, and these transactions may in turn generate option loss for other buyers. We therefore assess the aggregate importance of this mechanism by quantifying how much observed transaction activity can be attributed to the self-reinforcing cascades generated by option loss.

6.3.1 Framework

We adopt a binary treatment framework in which each buyer is either treated or untreated. A buyer becomes treated upon experiencing her first option loss event. Treatment shifts the buyer's daily purchase probability by Δp. Subsequent option loss events have no additional marginal effect. Under this assumption, a treated buyer purchases with daily probability $p_0 + \Delta p$ and an untreated buyer purchases with probability p_0, where p_0 is the counterfactual base rate absent the option-loss channel.

Averaging across all buyer-days, the observed equilibrium purchase probability satisfies:

$$p = p_0 + \sigma \cdot \Delta p, \tag{10}$$

where σ is the fraction of buyer-days affected by option loss and $\Delta p = 0.6$ (in percentage points) is the estimated causal effect from the baseline difference-in-differences specification (Table 2).

The counterfactual base rate and the cascade multiplier follow immediately:

$$p_0 = p - \sigma \cdot \Delta p, \qquad M = \frac{p}{p_0} = \frac{p}{p - \sigma \cdot \Delta p}. \tag{11}$$

All inputs are directly observed or estimated from the data: $p = 0.66$ and $\Delta p = 0.6$ (in percentage points). The remaining object, σ, depends on the assumed duration of the treatment effect[7].

[7]Under the additional assumption that the treatment stock is proportional to the aggregate purchase rate, i.e., $\sigma(p) = (\sigma/p) \cdot p$, the multiplier admits a feedback loop interpretation. Consider the economy starting from the base purchase rate p_0. In Round 0, base purchases generate a treatment stock of $\sigma_0 = (\sigma/p) \cdot p_0$. In Round 1, these treated buyer-days purchase at rate Δp above base, producing $\Delta p_1 = \sigma_0 \cdot \Delta p = (\sigma/p) \cdot \Delta p \cdot p_0$ additional purchases. In Round 2, the additional purchases Δp_1 raise the treatment stock by $\Delta\sigma_1 = (\sigma/p) \cdot \Delta p_1$, which generates $\Delta p_2 = \Delta\sigma_1 \cdot \Delta p = [(\sigma/p) \cdot \Delta p]^2 \cdot p_0$ further purchases. Continuing this logic, Round n contributes $\Delta p_n = [(\sigma/p) \cdot \Delta p]^n \cdot p_0$. Summing all rounds:

$$p = p_0 \sum_{n=0}^{\infty} \left(\frac{\sigma}{p} \cdot \Delta p \right)^n = \frac{p_0}{1 - \frac{\sigma}{p} \cdot \Delta p}.$$

Each round involves two links: purchases raise the treatment stock (scaled by σ/p) and treatment raises purchases (scaled by Δp). The product $(\sigma/p) \cdot \Delta p$ is the per-round propagation rate.

6.3.2 Results

Before presenting results, we assume that the treatment effects are persistent and permanent. The event study estimates in Figure 5 provide the supporting evidence that the treatment effect emerges immediately upon the first option loss and remains relatively stable over the subsequent days, suggesting that the permanent-effect assumption is a reasonable approximation, at least within the horizon of our data.

Every buyer-day in which the buyer has previously experienced one option loss carries the elevated purchase probability $p_0 + \Delta p$. The relevant measure of treatment exposure is the *stock*:

$$\sigma = 0.33,$$

where σ is the fraction of buyer-days in the panel in which the absorbing treatment indicator equals one. Table 9 reports the cascade quantification results. The cascade of option loss accounts for approximately 30% of observed purchases, amplifying the base transaction rate by a factor of 1.4. In the absence of the option-loss effects, the counterfactual daily transaction probability would decline from 0.66% to about 0.46%, implying a substantial reduction relative to the observed level.

The multiplier, 1.4, provides a quantitative measure of how strongly buyer-side spillovers amplify transaction activity in the Beijing housing market. The mechanism operates as follows: each transaction removes a property from the choice sets of other active searchers. In response, affected buyers revise upward their assessment of market tightness and accelerate their own purchase decisions, as documented in our empirical results. These additional transactions, in turn, generate option loss for yet another set of searchers, producing a self-reinforcing feedback cascade.

This feedback cascade is informative about the dynamics of the Beijing market during our sample period. The approximately 30% increase in transaction prices per square meter observed between August 2015 and July 2016 (Panel (d) in Figure 3) occurred alongside rising market tightness, as the ratio of transactions and visits to new listings increased steadily. Our results suggest that this tightening was partly self-reinforcing: as transactions accumulated, the resulting option-loss shocks compressed buyers' continuation values, accelerated subsequent purchases, and generated further signals of market tightness for remaining searchers. The cascade multiplier indicates that buyer congestion substantially amplified transaction activity beyond what would have occurred based on fundamentals alone.

Table 9: The Results of Cascade Quantification

	Parameters
Treatment exposure (σ)	0.33
Counterfactual purchase rate (p_0)	0.46
Cascade multiplier (M)	1.43
Share of p due to cascade ($\sigma \cdot \Delta p / p$)	30%

Notes: This table reports the counterfactual decomposition of the observed daily purchase probability (in percentage points) $p = 0.66$ into a base component p_0 and a cascade component $\sigma \cdot \Delta p$, where $\Delta p = 0.6$ is the estimated treatment effect of option loss.

7 Conclusion

In the search process of the housing market, buyers may be unable to perfectly recall previously inspected properties because those properties can be sold to other buyers before they make a decision. This paper examines whether such option loss affects buyers' subsequent search and purchase decisions, and whether individual transactions can generate transaction cascades through buyers' search process.

To understand how option loss affects search and purchase behavior of the buyer, we develop a sequential search model with recall, option loss, and Bayesian updating. Property sales serve a dual role: they remove options and convey information. Buyers update their posterior toward tighter market states upon observing a sale, which lowers the continuation value and stopping threshold, generating the immediate behavioral response documented empirically. The model yields sharp theoretical results on belief dynamics, value monotonicity, threshold stopping, and the triggering of recall purchases at sale shocks.

The empirical evidence supports these predictions. Using detailed search and transaction data from a major Chinese housing platform, we exploit the quasi-random timing of sales of previously inspected properties in a staggered difference-in-differences design. We find that option loss increases the daily purchase probability of affected buyers by approximately 67%. This result is robust to alternative specifications, including duration models that account for the stopping nature of the purchase decision. Consistent with the signal-based mechanism, the effect is larger in tighter markets and among buyers with larger choice sets. Heterogeneity analyses further show that the response is weaker among buyers with higher price preferences, but stronger among mortgage buyers and when the lost property is relatively inexpensive or falls within the buyer's preferred price range.

Option loss also changes the way buyers search. After losing a previously in-

spected option, buyers reduce the number of daily visits by approximately 0.04, mainly through fewer visits to new properties. Buyers also broaden their search criteria and become more likely to visit properties whose prices and sizes exceed their previously stated preferences. Finally, option loss is associated with higher transaction prices: affected buyers purchase properties at prices approximately 0.7 percent to 1.3 percent higher on average. These findings suggest that option loss not only accelerates purchase timing, but also weakens buyers' bargaining position or leads them to relax budget constraints.

The results have broader implications for housing market dynamics. Our quantification suggests that transaction cascades generated by option loss account for about 30 percent of observed transaction activity, with an implied multiplier of 1.43. This buyer-side propagation mechanism complements existing explanations of housing booms based on credit conditions, supply constraints, and market liquidity. More broadly, the findings show that transactions have informational and behavioral effects beyond the parties directly involved. By removing options and reshaping beliefs for other active searchers, individual sales can amplify market activity and contribute to self-reinforcing housing booms.

These findings also have welfare implications. Option loss appears to redistribute surplus from buyers to sellers: affected buyers transact more quickly, search less, and pay higher prices. This suggests that dynamic market environments can create search frictions even when buyers remember previously inspected properties, because market exit can make recall infeasible. Understanding this form of imperfect recall is therefore important for explaining not only individual search behavior, but also aggregate transaction activity and price dynamics in housing markets.

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Online Appendix

A Robustness Check of the Empirical Results

Duration analysis. We therefore complement the difference-in-differences analysis with a duration model. The duration framework treats purchase as a stopping event and estimates the conditional hazard of purchase among buyers who are still searching. Specifically, it asks whether option loss raises the probability of purchase on day (t), conditional on the buyer not having purchased before day (t). This approach accounts for duration dependence in purchase hazards and separates the effect of option loss from the mechanical requirement that treated buyers remain unpurchased until the treatment occurs.

We estimate a duration model similar to de Wit and van der Klaauw [2013] using an extended Cox proportional hazard specification on a buyer-day panel. The dependent variable is the purchase hazard, defined as the conditional probability that buyer i purchases a house at duration t, given that the buyer has not purchased previously. The baseline hazard consists of three components. First, $\lambda(t)$ captures duration dependence in search duration. Second, $\psi_n(n_i)$ controls for neighborhood-specific market conditions. Third, $\psi_d(d_t)$ captures aggregate calendar-day effects.

The key explanatory variable is the time-varying indicator function $\mathbf{1}(\tau_{pi} \geq t)$, where τ_{pi} denotes the timing of buyer i's option-loss event. The indicator equals one after the option loss occurs and zero otherwise. The coefficient δ therefore captures how option loss shifts the purchase hazard conditional on the buyer remaining unmatched up to duration t. The vector X_i includes additional buyer characteristics, including gender, mortgage usage, and preferences over price, number of rooms, and housing size. The estimated specification is as follow:

$$\theta_i(t \mid X_i, n_i, w_t, \tau_{pi}) = \lambda(t)\psi_n(n_i)\psi_w(w_t) \exp\left[X_i'\gamma + \delta \cdot \mathbf{1}(\tau_{pi} \geq t)\right]. \tag{A1}$$

Abbring and Van Den Berg [2003] demonstrate that identification of this type of duration model relies on a no-anticipation assumption. Specifically, conditional on observed buyer characteristics, neighborhood conditions, and calendar-time effects, the buyer's purchase hazard prior to the option-loss event does not depend on the exact future timing of the option loss. The identifying variation therefore comes from the timing of option-loss events rather than from time-invariant differences between buyers who experience option loss and those who do not.

Table A2 reports estimates from Cox proportional hazard models examining the effect of option loss on the likelihood of purchase. Across all specifications, option loss significantly increases the purchase hazard. In our preferred specification,

reported in Column (3), the estimated hazard ratio is 1.6, implying that option loss increases the purchase hazard by a factor of 1.6. This estimate is highly consistent with the baseline result in Table 2, where option loss increases the daily purchase probability by approximately 67%. The similarity between the Cox estimates and the baseline difference-in-differences estimates suggests that our findings are not driven by the specific functional form or model specification used to estimate purchase responses.

Diagnostics for heterogeneous treatment effects. Recent advances in the DID literature have emphasized that, under staggered treatment adoption, the conventional two-way fixed effects (TWFE) estimator may produce biased estimates when treatment effects are heterogeneous across cohorts or over time [Callaway and Sant'Anna, 2021, de Chaisemartin and D'Haultfœuille, 2020, Goodman-Bacon, 2021]. In particular, the TWFE estimator can be expressed as a weighted average of cohort-specific treatment effects, where some weights may be negative. Such negative weights can distort the overall estimate and potentially lead to misleading conclusions.

To assess the relevance of this issue in our setting, we first examine the implicit weighting structure of the TWFE estimator using the `twowayfeweights` procedure. The results in Table A3 and Figure A3 indicate that negative weights are negligible in magnitude and account for only a small fraction of the total weights, suggesting that the baseline estimates are unlikely to be driven by pathological weighting.

As a further robustness check, we implement the stacked DID approach following Sun and Abraham [2021], using the `eventstudyinteract` command. This method constructs cohort-specific comparison samples and avoids the negative weighting problem by design. Due to the substantial computational burden, estimating the model on the full sample becomes extremely time-consuming. We therefore randomly draw a 10% subsample for estimation. The results in Table A4 show that the estimated treatment effect is approximately 0.7 (Column (4)), which is similar in magnitude to our baseline estimate of 0.6. Figure A4 presents the event study plots using Sun and Abraham [2021] estimator with 10% random sample at week level. The results continue to support the parallel-trends assumption, and the post-treatment effect remains clearly visible. This consistency provides strong evidence that our findings are not driven by heterogeneous treatment effects or the negative weight issue inherent in the TWFE estimator.

Purchase probability conditional on visits. In the baseline specification, we impute the transaction probability to zero for days without any recorded visits. While this approach is consistent with the economic intuition that transactions cannot occur in the absence of visits, it may introduce bias if the absence of visits is itself

endogenous or systematically related to treatment. To address this concern, we conduct a robustness check using a non-imputed sample that retains only observations with at least one visit. This also allows to decompose purchase probability into the product of the number of visits and the purchase probability conditional on visits. We then re-estimate the model on this restricted sample. The event study results, reported in Figure A5, remain qualitatively robust. In particular, we continue to observe an immediate increase in the conditional transaction probability following the occurrence of option loss.

Notably, the estimated treatment effects are substantially larger in magnitude than those in the baseline specification. In Table A5, the estimated effects are between 1.8 and 3.8 , much larger than the baseline estimates between 0.6 and 1 . This is expected, as the estimates in this exercise should be interpreted as the purchase probability conditional on visits, rather than unconditional purchase probabilities. This implies that option loss not only increases buyers' overall purchase probability, but also significantly raises the likelihood of purchase conditional on each visit, with an even larger magnitude. The increase in the conditional purchase probability is larger than that of the overall purchase probability, implying a decline in visits after option loss. The rise in purchase probability is thus primarily driven by the conditional component. Together, these findings suggest that our main results are not driven by the imputation procedure and remain robust under alternative sample constructions.

Permutation test. To further assess whether our baseline estimates are driven by spurious correlations, we conduct a permutation test. Specifically, we randomly select the same number of treated searchers as in the original sample and randomly assign them placebo treatment timing drawn from the empirical distribution of actual treatment timing. Based on these randomly assigned treatment statuses and timing, we construct a placebo treatment indicator following the same definition as in the baseline specification.

We then re-estimate the baseline regression using this placebo treatment variable and repeat the procedure 500 times. This generates a distribution of placebo estimates that captures the extent of spurious treatment effects under random assignment, as well as a distribution of the corresponding p-values. Figure A6 presents the results. Comparing the actual estimate to these distributions allows us to evaluate whether the observed effect is statistically distinguishable from those obtained under random treatment assignment.

The distribution of placebo estimates is tightly centered around zero, and all of the estimates are statistically insignificant. In contrast, the preferred estimates in Column (2) of Table 2 is around 0.64 and lies far in the right tail of the placebo distribution. Consistent with this pattern, the distribution of p-values from the

placebo regressions is concentrated at high values, indicating a lack of statistical significance under random assignment. Taken together, these results suggest that the baseline estimate is unlikely to be driven by chance and provides strong support for a genuine treatment effect.

B Additional Figures

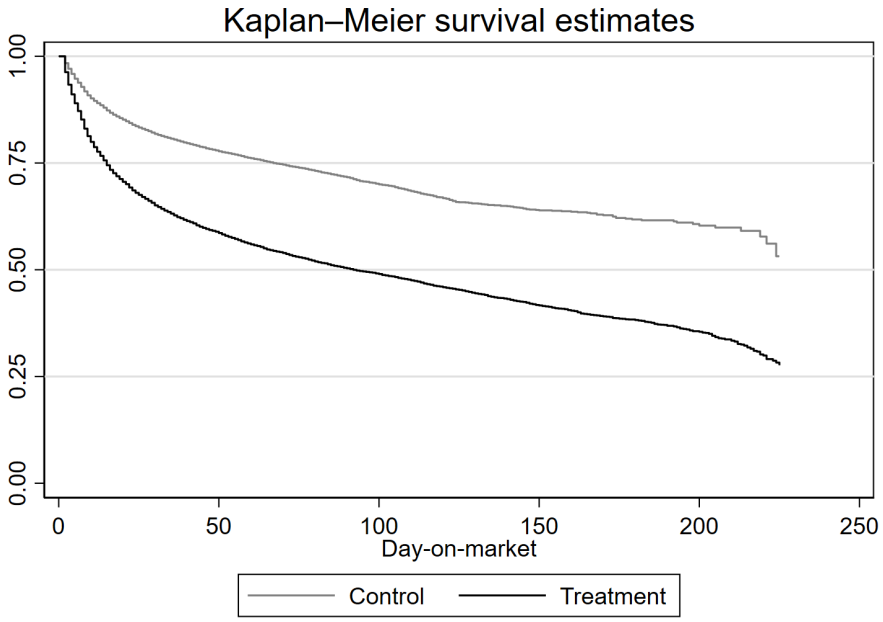


Figure A1: Kaplan–Meier Estimates of Purchase Timing

Notes: The figure presents Kaplan–Meier survival estimates of purchase timing for treated and control buyers. The dependent variable is the probability that a buyer remains unmatched over time. The treatment group consists of buyers who experience *Option Loss*, defined as a previously visited property being purchased by another buyer, while the control group consists of buyers who do not experience option loss. The horizontal axis reports search duration measured in days since the buyer's first observed visit. Since the observation window is finite, buyers who do not purchase within the sample period are treated as right-censored observations.

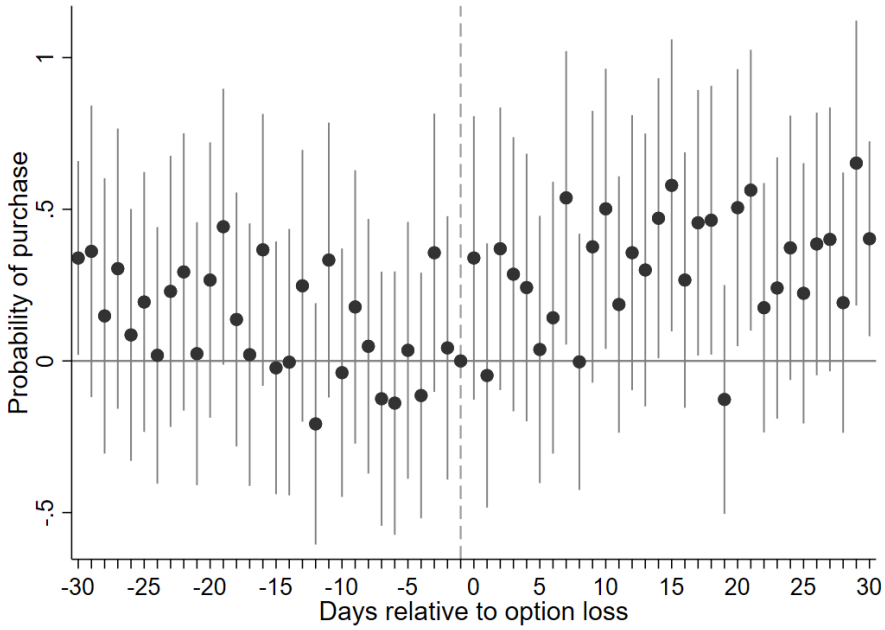


Figure A2: Dynamic Effects of Sales of Unvisited Neighboring Properties

Notes: This figure reports event-study estimates using the sale of an unvisited neighboring property as the placebo treatment event. Neighboring properties are defined as properties listed in the same week and located in the same apartment complex as the property generating the actual option-loss event. The period immediately preceding the placebo sale is omitted as the reference period. The dots report coefficient estimates and the bars indicate 95% confidence intervals.

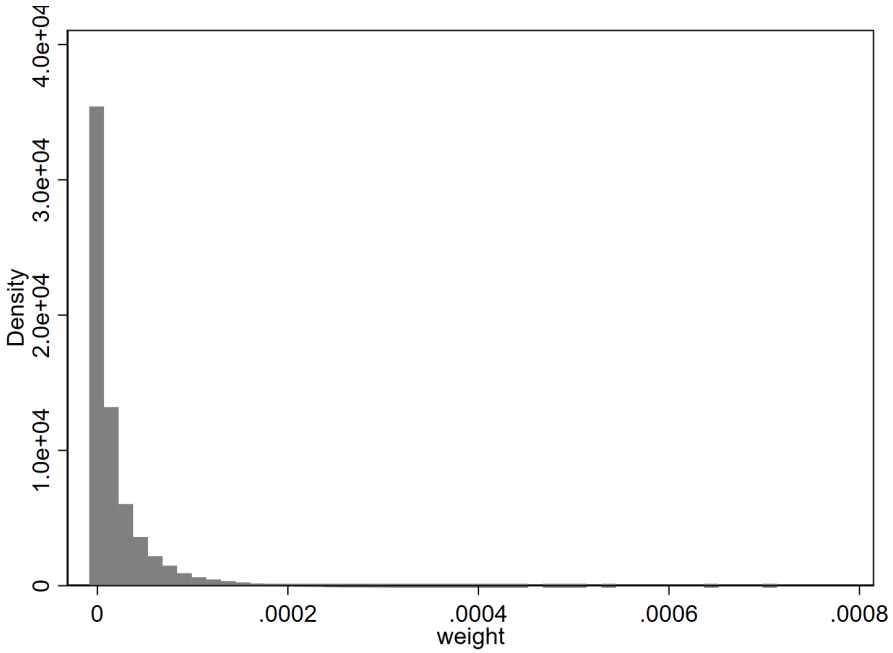


Figure A3: Distribution of Two-Way Fixed Effects Treatment Weights

Notes: This figure plots the distribution of weights associated with each treatment comparison, computed using the `twowayfeweights` command in Stata. The distribution shows that almost all weights are positive, with only a very small share of negative weights.

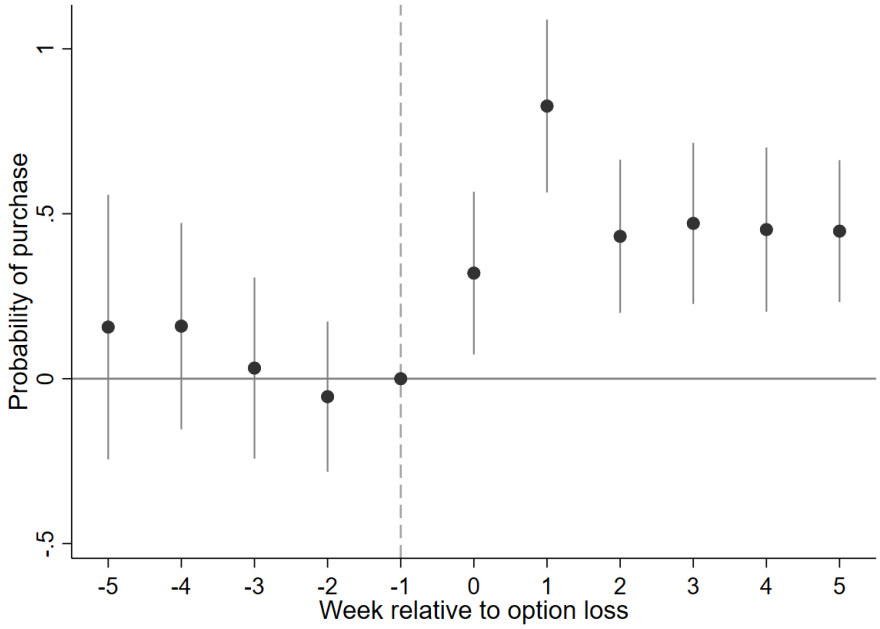


Figure A4: The Dynamic Effects on Purchase Probability Using Sun and Abraham [2021] Estimator

Notes: This figure reports event-study estimates based on the interaction-weighted estimator proposed by Sun and Abraham [2021], which accounts for treatment effect heterogeneity in staggered adoption settings. Due to the computational burden of implementing this estimator, we use a 10% random sample of the baseline sample and estimate the specification at the weekly level. The event window covers five weeks before and after option loss. Observations outside this window are binned into the -5 and 5 week endpoints. Vertical bars represent 95 percent confidence intervals.

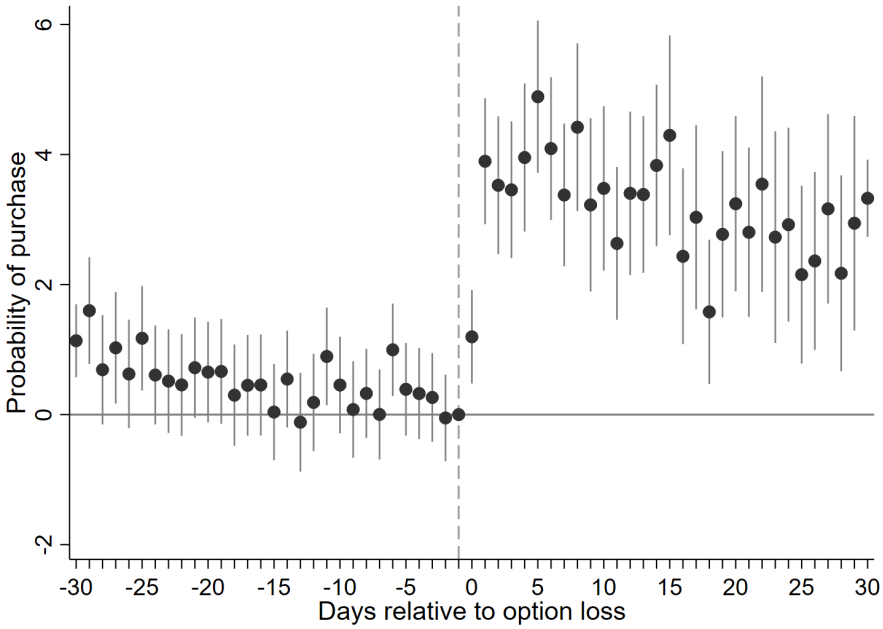
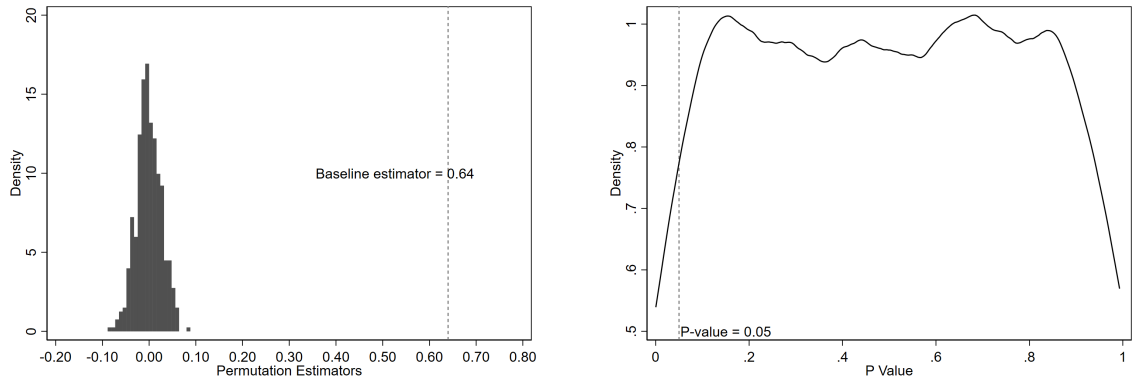


Figure A5: The Dynamic Effects on Purchase Probability Conditional on Visits

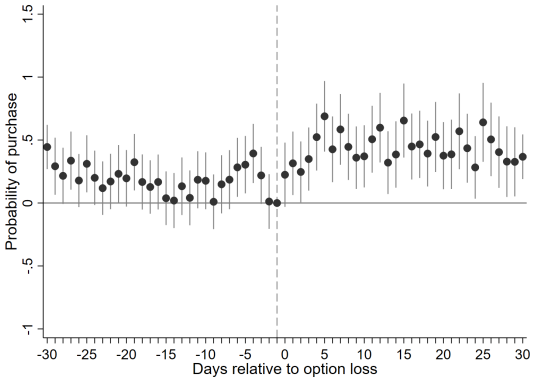
Notes: This figure reports the event-study estimates of the effect of option loss on the conditional transaction probability. Unlike the baseline specification, the sample is restricted to buyer-day observations with at least one recorded visit, and days without visits are not imputed as zero transaction days. The dependent variable equals 100 if the buyer completes a transaction on a day with at least one visit. Vertical bars represent 95 percent confidence intervals.



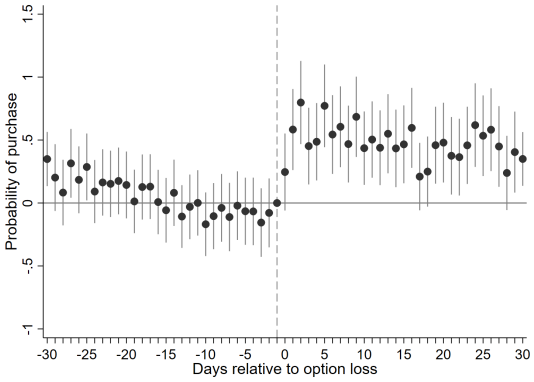
(a) Distribution of Permutation Estimators (b) Distribution of Permutation P Value

Figure A6: Results of permutation test

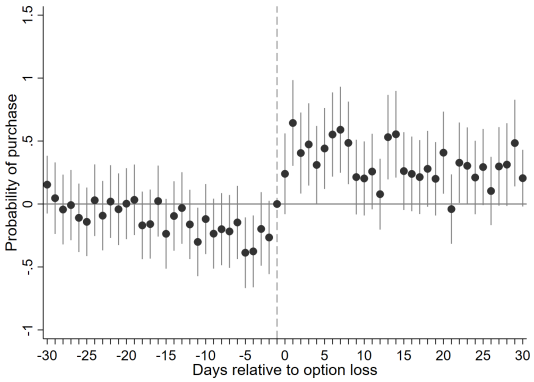
Notes: This figure reports the results of the permutation test based on 500 random assignments of treatment status. Panel (a) plots the histogram of placebo treatment estimates from the 500 replications using specification in Column (2) of Table 2. The vertical line indicates the actual baseline estimate of 0.64. Panel (b) plots the kernel density of the corresponding placebo p-values. The dashed vertical line indicates the 95 percent significance level. The vast majority of placebo estimates have p-values above 0.05, suggesting that the baseline effect is unlikely to be generated by random treatment assignment.



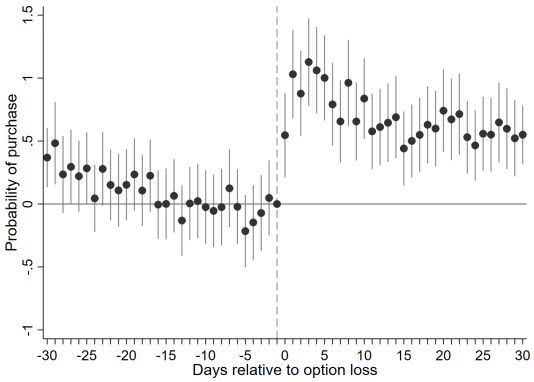
(a) First Quartile of Market Tightness



(b) Second Quartile of Market Tightness



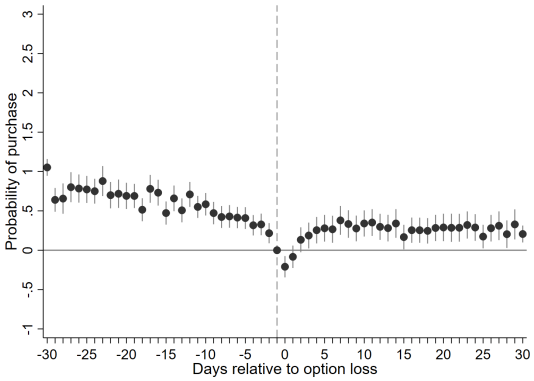
(c) Third Quartile of Market Tightness



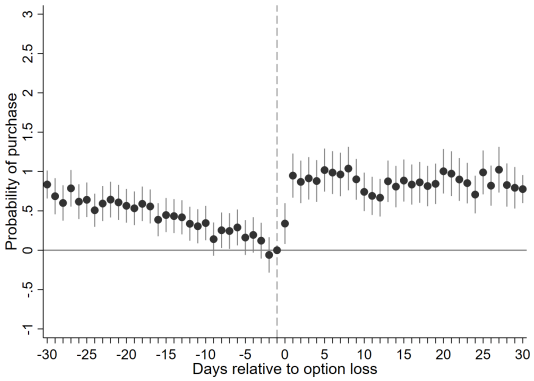
(d) Fourth Quartile of Market Tightness

Figure A7: Heterogeneous Effects of Option Loss by Market Tightness

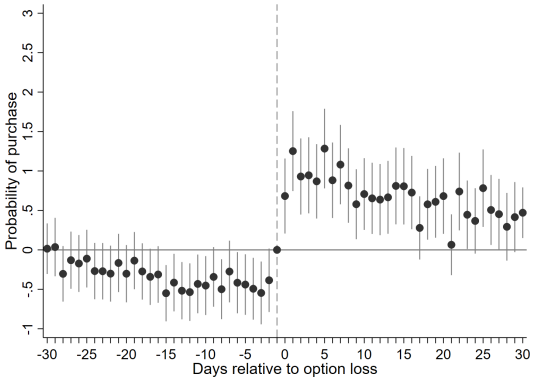
Notes: This figure reports the estimated effects of option loss separately by quartiles of market tightness. Market tightness is measured at the neighborhood-week level in the week when option loss occurs and is defined as the ratio of transaction volume (in thousands) to the total number of listed properties. The quartile cutoffs are 13.45, 20.24, and 29.20. In each panel, treated buyers in the corresponding market-tightness quartile are compared with the control group of buyers who have not yet experienced option loss.



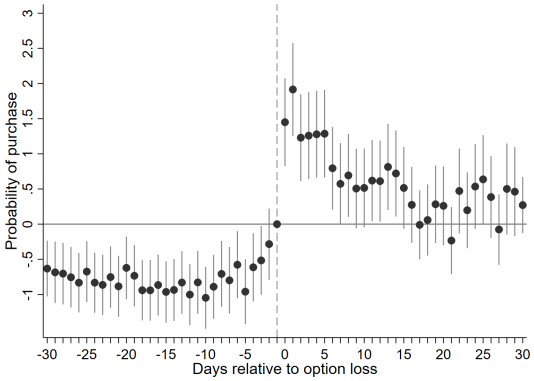
(a) First Quartile of Choice Set Size



(b) Second Quartile of Choice Set Size



(c) Third Quartile of Choice Set Size



(d) Fourth Quartile of Choice Set Size

Figure A8: Heterogeneous Effects of Option Loss by Choice Set Size

Notes: This figure reports the estimated effects of option loss separately by quartiles of choice set size. Choice set size is measured by the number of properties a buyer had inspected before experiencing option loss. The quartile cutoffs are 2, 4, and 6. In each panel, treated buyers in the corresponding quartile are compared with the control group of buyers who have not yet experienced option loss.

C Additional Tables

Table A1: Balance Test for Buyer Characteristics by Treatment Status

	Control Group		Treatment Group		Difference
Variable	Obs.	Mean	Obs.	Mean	Treated − Control
Gender	104,974	0.510	46,712	0.511	0.001
Mortgage	104,974	0.819	46,712	0.835	0.015***
Search duration	104,974	17.4	46,712	45.2	27.9***
Neighborhood	104,974	120.5	46,712	123.2	2.8***
Upper stated price	104,974	441.3	46,712	434.6	-6.7***
Lower stated price	104,974	353.3	46,712	347.6	-5.6***
Upper stated size	104,974	105.2	46,712	102.5	-2.7***
Lower stated size	104,974	81.8	46,712	79.6	-2.3***
Upper stated rooms	104,974	2.466	46,712	2.447	-0.019***
Lower stated rooms	104,974	2.011	46,712	1.985	-0.027***

Notes: This table reports balance tests between the treatment and control groups. The last column reports the difference in means between the treatment and control groups. Search duration is measured as the number of days between a buyer's first and last recorded property visit. Neighborhood is the numerical code of the neighborhood in which the buyer searches; its mean reflects differences in the composition of neighborhoods across groups and has no direct economic interpretation. Significance levels are based on two-sample t-tests. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Table A2: Effects of Option Loss on Purchase Hazard using Cox Model

	Purchase Hazard (Cox Model)		
	(1)	(2)	(3)
Option Loss	1.973***	1.970***	1.611***
	(0.028)	(0.027)	(0.032)
# visits		1.543***	1.600***
		(0.006)	(0.017)
N	4,008,625	4,008,625	4,008,625
Buyer characteristics	✓	✓	✓
Neighborhood strata			✓
Day strata			✓

Notes: Standard errors are clustered at the searcher level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. The table reports hazard ratios from extended Cox proportional hazard models estimated on a buyer-day panel. The dependent variable is the purchase hazard. The key explanatory variable, *Option Loss*, is a time-varying indicator equal to one after a previously visited property has been purchased by another buyer, and zero otherwise. All specifications control for buyer characteristics, including gender, mortgage usage, and preferences over price, number of rooms, and housing size. Columns (2)–(3) additionally control for buyer search intensity measured by the number of visits. Column (3) allows the baseline hazard to vary across neighborhoods and calendar days.

Table A3: Decomposition of ATT Weights

	# ATTs	Sum of weights
Positive weights	43,602	1.0104
Negative weights	9,584	-0.0104
Total	53,186	1.0000

Notes: This table reports the decomposition of weights in the two-way fixed effects estimator. Under the parallel trends assumption, the estimated coefficient can be expressed as a weighted average of 53,186 cohort-time average treatment effects (ATTs). Among these, 43,602 receive positive weights, while 9,584 receive negative weights.

Table A4: Effects of Option Loss on Purchase Probability Using Sun and Abraham [2021] Estimator

	Purchase			
	(1)	(2)	(3)	(4)
Option Loss	1.126***	0.719***	0.721***	0.698***
	(0.068)	(0.068)	(0.068)	(0.080)
N	393,420	393,420	393,420	382,371
R^2	0.153	0.159	0.159	0.322
Searcher F.E.	✓	✓	✓	✓
Day F.E.	✓	✓	✓	✓
Day-on-market F.E.		✓	✓	✓
Visits			✓	✓
Payment × Day F.E.				✓
Gender × Day F.E.				✓
Neighbor × Day F.E.				✓
Group × Day F.E.				✓

Notes: Standard errors are clustered at the searcher level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. The dependent variable is the purchase probability at the day level. This table uses the interaction-weighted estimator proposed by Sun and Abraham [2021] to account for treatment effect heterogeneity in staggered adoption settings. Due to the computational burden of implementing this estimator, we use a 10% random sample of the baseline sample. The key explanatory variable, *Option Loss*, indicates that a property previously visited by the searcher has been purchased by another buyer, resulting in the loss of an option from the searcher's choice set. Column (2) additionally controls for day-on-market fixed effects. Column (3) further controls for buyer search intensity measured by the number of visits. Column (4) additionally includes interactions between day fixed effects and searcher characteristics, including payment method, gender, neighborhood preference, and preferred price group. All specifications include searcher and day fixed effects.

Table A5: Effects of Option Loss on Purchase Probability Conditional on Visits

	Purchase			
	(1)	(2)	(3)	(4)
Option Loss	3.819***	2.610***	2.190***	1.857***
	(0.105)	(0.106)	(0.106)	(0.119)
# visits			-1.333***	-1.313***
			(0.032)	(0.035)
N	413,065	413,065	413,065	395,937
R^2	0.372	0.382	0.387	0.524
Searcher F.E.	✓	✓	✓	✓
Day F.E.	✓	✓	✓	✓
Day-on-market F.E.		✓	✓	✓
Payment × Day F.E.				✓
Gender × Day F.E.				✓
Neighbor × Day F.E.				✓
Group × Day F.E.				✓

Notes: Standard errors are clustered at the searcher level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. The dependent variable is the purchase probability at day level. The key explanatory variable, *Option Loss*, indicates that a property previously visited by the searcher has been purchased by another buyer, resulting in the loss of an option from the searcher's choice set. Columns (2)–(4) additionally control for Day-on-market fixed effects, where day-on-market is defined as the number of days elapsed since the searcher's first visit. Columns (3)–(4) additionally control for the number of visits at day level. Column (4) includes interactions between day fixed effects and searcher characteristics, including payment method, gender, neighborhood preference, and preferred price group. All specifications include searcher and week fixed effects. This regression is conducted at the buyer-day level and restricts the sample to observations in which the buyer makes at least one visit, thereby excluding weeks with no visits.

D Proofs for Section 2

Derivation of the closed-form threshold. Setting $G(b^*, m) = 0$ in (6) and multiplying through by 2 yields the quadratic

$$(1-m)\,b^{*2} \;-\; 2\,b^* \;+\; (1-2\kappa) \;=\; 0.$$

Applying the quadratic formula gives two roots; the larger root exceeds $1/(1-m) \geq 1$ and lies outside the unit interval, while the smaller root is the economically meaningful stopping threshold,

$$b^*(m) \;=\; \frac{1-\sqrt{1-(1-m)(1-2\kappa)}}{1-m}. \tag{A2}$$

To verify $b^*(m) \in (0,1)$, let $D(m) \equiv 1-(1-m)(1-2\kappa)$. For $m \in [0,1)$ and $\kappa \in (0, \frac{1}{2})$, $(1-m)(1-2\kappa) \in (0,1)$, so $D(m) \in (0,1)$ and $\sqrt{D(m)} < 1$. Hence the numerator of (A2) is positive and $b^*(m) > 0$. For the upper bound, $b^*(m) < 1$ if and only if $\sqrt{D(m)} > m$, equivalently $D(m) > m^2$. Direct expansion gives

$$D(m) - m^2 \;=\; 2\kappa(1-m) + m(1-m) \;=\; (1-m)(2\kappa+m) \;>\; 0,$$

so $b^*(m) < 1$. The boundary values are $b^*(0) = 1-\sqrt{2\kappa}$ and $\lim_{m\to 1} b^*(m) = \frac{1}{2} - \kappa$. The latter limit reflects that when house b is certain to vanish, continuing yields only $E[x] - \kappa = \frac{1}{2} - \kappa$, and the buyer stops if and only if b exceeds this continuation value.

Proof of Proposition 2. Differentiating (6), $\partial G/\partial b = (1-m)b - 1 \leq -m \leq 0$ for $b \in [0,1]$, so $G(\cdot, m)$ is weakly decreasing (strictly so at every interior b). At the boundaries, $G(0,m) = \frac{1}{2} - \kappa > 0$ since $\kappa < \frac{1}{2}$, and $G(1,m) = -\frac{m}{2} - \kappa < 0$. Continuity and strict monotonicity imply a unique interior root $b^*(m) \in (0,1)$, and $G(b,m) \leq 0 \iff b \geq b^*(m)$. □

Proof of Proposition 1. From (2),

$$\pi_+ - \pi_0 \;=\; \frac{\pi_0(1-\pi_0)\,(h_T - h_L)}{\pi_0\,h_T + (1-\pi_0)\,h_L} \;>\; 0,$$

and similarly from (3),

$$\pi_0 - \pi_- \;=\; \frac{\pi_0(1-\pi_0)\,(h_T - h_L)}{\pi_0\,(1-h_T) + (1-\pi_0)\,(1-h_L)} \;>\; 0.$$

Strict monotonicity of $m(\pi)$ in π gives the corresponding inequalities on m. □

Proof of Proposition 3. Differentiate the closed-form expression (A2) directly. Let $D(m) \equiv 1-(1-m)(1-2\kappa)$, so $D'(m) = 1-2\kappa > 0$ and $b^*(m) = [1-\sqrt{D(m)}]/(1-m)$. The quotient rule gives

$$\frac{db^*}{dm} \;=\; \frac{1}{(1-m)^2}\left[-\frac{(1-2\kappa)(1-m)}{2\sqrt{D(m)}} + 1 - \sqrt{D(m)}\right] \;=\; \frac{(1-m)(1-2\kappa) - 2 + 2\sqrt{D(m)}}{2(1-m)^2\sqrt{D(m)}}.$$

Using $(1-m)(1-2\kappa) = 1 - D(m)$, the numerator simplifies to

$$\big[1 - D(m)\big] - 2 + 2\sqrt{D(m)} \;=\; -1 + 2\sqrt{D(m)} - D(m) \;=\; -\big(\sqrt{D(m)} - 1\big)^2,$$

which is the negative of a square and hence strictly negative whenever $D(m) \neq 1$, i.e. whenever $\kappa > 0$. Therefore

$$\frac{db^*}{dm} \;=\; -\frac{\big(\sqrt{D(m)} - 1\big)^2}{2(1-m)^2\sqrt{D(m)}} \;<\; 0.$$

□

Proof of Proposition 4. By Proposition 1, $m(\pi_+) > m(\pi_-)$, and by Proposition 3, $b^*_+ < b^*_-$. By the threshold characterization in Proposition 2, $G(b, m(\pi_-)) > 0$ for $b < b^*_-$ (the buyer continues after $\overline{S}$) and $G(b, m(\pi_+)) \leq 0$ for $b \geq b^*_+$ (the buyer stops after S). The intersection of these regions is $[b^*_+, b^*_-)$. With $b \sim U[0,1]$, the stopping probability under threshold b^* is $\Pr(b \geq b^*) = 1 - b^*$, so the change in stopping probability between S and $\overline{S}$ is $\Delta P = (1 - b^*_+) - (1 - b^*_-) = b^*_- - b^*_+$. □

E Full Continuous-Time Model

This appendix develops the continuous-time version of the search model introduced in Section 2. The environment enriches the three-period model in two ways. First, market tightness takes one of two states, and the buyer updates her belief about the state from sales observed among previously inspected houses. Second, houses are ex-ante heterogeneous and the buyer inspects them in descending order of pre-visit quality, so that the value of newly arriving options deteriorates as search progresses. This second feature is the key departure from an i.i.d. arrival model. It makes the recalled option genuinely valuable in the continuation region: a buyer holding a strong recalled house may stop and exercise it because the houses she would inspect next are worse, not only because an incoming draw happens to dominate it.

The appendix proceeds in four steps. Section E.2 establishes the belief dynamics: a sale shock raises the perceived hazard, while the absence of sales lowers it. Section E.3 studies the buyer's post-update stopping and search problem at a fixed perceived hazard m. At each information event, the buyer solves this problem given the current perceived hazard; the comparative statics in m then characterize how sale shocks affect stopping after beliefs are updated. Sections E.4–E.7 characterize the threshold rule and the effect of sale shocks. Section E.8 collects implications for choice-set size, visits, and transaction prices.

E.1 Environment

Houses and pre-visit signals. Each house is characterized, from the perspective of a given buyer, by a pre-visit signal $s \in [0, \bar{s}_0]$. The signal summarizes the buyer's pre-visit assessment of the property based on listing-level information observable before a visit, such as neighborhood, asking price, floor area, number of rooms, and photographs. Conditional on s, the buyer's idiosyncratic match surplus from the house is

$$x \sim F(\cdot \mid s),$$

where $F(\cdot \mid s)$ is a continuous CDF on $[0, \bar{x}]$ with density $f(\cdot \mid s)$ strictly positive on $(0, \bar{x})$ for each $s > 0$. Both s and x are buyer-specific: a property that is attractive to one buyer need not be equally attractive to other buyers.

Directed search and the queue position. The buyer inspects houses in descending order of pre-visit signal. Let $\bar{s}$ denote the signal of the next house she would inspect. After each inspection, the queue position falls deterministically:

$$\bar{s}' = \phi(\bar{s}) < \bar{s},$$

where ϕ is continuous, strictly increasing, $\phi(0) = 0$, and

$$\phi^K(\bar{s}) \to 0 \qquad \text{as } K \to \infty$$

for every $\bar{s} \in (0, \bar{s}_0]$. Between inspections, the queue position is constant. The buyer does not move down the queue merely because clock time passes; she moves down the queue only when she inspects a new house.

This directed-search structure is a reduced-form analogue of the optimal-ordering rule in Weitzman [1979], under which alternatives are inspected in descending order of reservation-value indices. In our environment, previously inspected houses may be sold, beliefs update over market tightness, and the queue position interacts with the stopping threshold. We therefore impose descending-order search directly and study its consequences for option loss.

Inspection technology. Time is continuous, $t \geq 0$. Inspection opportunities arrive according to a Poisson process with rate $\lambda > 0$. While active, the buyer pays flow search cost $\kappa > 0$ and discounts future payoffs at rate $\rho > 0$.

Recall and option loss. The buyer retains her best previously inspected house, summarized by its surplus

$$b \in [0, \bar{x}].$$

At any time she may purchase this recalled house and terminate search. Each previously inspected house may also be sold to another buyer. If the sold house is the buyer's best recalled option, the recall state falls from b to 0.[8]

Two-state market tightness. The market is in one of two states,

$$\theta \in \{L, T\},$$

where L denotes a loose market and T denotes a tight market. Sale hazards are state dependent:

$$0 \leq h_L < h_T < \infty.$$

The buyer does not observe θ. She holds posterior belief

$$\pi = \Pr(\theta = T) \in [0, 1].$$

The posterior mean option-loss hazard is

$$m(\pi) = \pi h_T + (1 - \pi) h_L \in [h_L, h_T]. \tag{D.1}$$

[8]For tractability, and without loss of the main economic insight, we do not explicitly model the buyer's second- or third-best recalled options. Doing so would require tracking the full set of previously visited properties and their availability over time, substantially complicating the state space.

Because there are only two states, m is a sufficient statistic for the posterior:

$$\pi = \frac{m - h_L}{h_T - h_L}. \tag{D.2}$$

Thus knowing m is equivalent to knowing the posterior π, and all belief-dependent objects can be written as functions of m.

The buyer monitors $N \geq 1$ previously inspected houses, including the best recalled house. Conditional on state θ, sales among monitored houses arrive at intensity Nh_θ. The number N affects the intensity of belief-relevant sale signals and the probability that a sale removes the buyer's best recalled option. It does not enter the post-update stopping problem directly: once beliefs have been updated, the buyer's payoff-relevant state is summarized by $(b, m, \bar{s})$.

E.2 Belief Dynamics

We first characterize how the perceived hazard responds to an observed sale and to an interval with no observed sale.

Proposition D.1 (A sale shock raises the perceived hazard)**.** *Suppose the buyer observes a sale among her monitored houses when her prior belief is $\pi \in (0,1)$. Then the posterior is*

$$\pi^+ = \frac{\pi h_T}{\pi h_T + (1-\pi)h_L}, \tag{D.3}$$

and the perceived hazard rises:

$$m(\pi^+) > m(\pi).$$

Proof. The likelihood of observing a monitored sale is proportional to Nh_θ. Since N is common across states, the likelihood ratio is determined by h_θ. Bayes' rule gives (D.3). Subtracting π,

$$\pi^+ - \pi = \frac{\pi h_T - \pi[\pi h_T + (1-\pi)h_L]}{\pi h_T + (1-\pi)h_L} = \frac{\pi(1-\pi)(h_T - h_L)}{\pi h_T + (1-\pi)h_L} > 0.$$

Since $m(\pi) = \pi h_T + (1-\pi)h_L$ is strictly increasing in π, it follows that $m(\pi^+) > m(\pi)$. □

Proposition D.2 (No sale lowers the perceived hazard)**.** *Suppose no monitored sale occurs over an interval of length $\Delta > 0$ when the belief at the start of the interval is $\pi \in (0,1)$. Then the posterior is*

$$\pi_\Delta = \frac{\pi e^{-Nh_T\Delta}}{\pi e^{-Nh_T\Delta} + (1-\pi)e^{-Nh_L\Delta}}, \tag{D.4}$$

and the perceived hazard falls:

$$m(\pi_\Delta) < m(\pi).$$

Proof. Conditional on state θ, the probability of no monitored sale over an interval of length Δ is $\exp(-Nh_\theta\Delta)$. Bayes' rule gives (D.4). Divide numerator and denominator by $e^{-Nh_L\Delta}$ and define

$$r = e^{-N(h_T-h_L)\Delta} \in (0,1).$$

Then

$$\pi_\Delta = \frac{\pi r}{\pi r + (1-\pi)}.$$

Hence

$$\pi_\Delta - \pi = \frac{\pi r - \pi[\pi r + (1-\pi)]}{\pi r + (1-\pi)} = \frac{\pi(1-\pi)(r-1)}{\pi r + (1-\pi)} < 0.$$

Since $m(\pi)$ is strictly increasing in π, $m(\pi_\Delta) < m(\pi)$. □

Sales occur more frequently in the tight state, so observing a sale shifts the posterior toward T and raises the perceived hazard. Conversely, an interval with no sales shifts the posterior toward L and lowers the perceived hazard. The factor N affects the speed of no-sale learning in (D.4), but not the direction of belief updating.

E.3 The Buyer's Post-Update Problem

We now formulate the buyer's stopping and search problem at a fixed perceived hazard m. This is not a full filtering HJB with the continuous drift of beliefs written explicitly. Instead, it is the post-update stopping/search problem the buyer solves given the current perceived hazard. Sale shocks affect behavior through the comparative statics of this problem as beliefs move from m^- to m^+.

Let $V_m(b,\bar{s})$ denote the value of a buyer with recalled surplus b, perceived hazard m, and current queue position $\bar{s}$. Immediate purchase gives payoff b. If the buyer continues, she pays flow cost κ. An inspection arrives at rate λ; upon inspection, the buyer observes $x \sim F(\cdot \mid \bar{s})$, may purchase the new house, and if she continues her next queue position becomes $\phi(\bar{s})$. The recalled option is lost at perceived hazard m, in which case the recall state becomes zero.

The value function satisfies

$$V_m(b,\bar{s}) = \max\{b, C_m(b,\bar{s})\}, \tag{D.5}$$

where

$$C_m(b,\bar{s}) = \frac{-\kappa + \lambda \mathbb{E}_{x|\bar{s}}\left[\max\left\{b,\, x,\, V_m(\max\{b,x\},\phi(\bar{s}))\right\}\right] + mV_m(0,\bar{s})}{\rho + \lambda + m}. \tag{D.6}$$

Equivalently, define the global Bellman operator $\mathcal{T}_m$ on bounded functions $W : [0,\bar{x}] \times [0,\bar{s}_0] \to \mathbb{R}$ by

$$(\mathcal{T}_m W)(b,\bar{s}) = \max\left\{b,\, \frac{-\kappa + \lambda \mathbb{E}_{x|\bar{s}}\left[\max\left\{b,\, x,\, W(\max\{b,x\},\phi(\bar{s}))\right\}\right] + mW(0,\bar{s})}{\rho + \lambda + m}\right\}. \tag{D.7}$$

The value function is the unique fixed point:

$$V_m = \mathcal{T}_m V_m.$$

At the boundary $\bar{s} = 0$, the unexplored pool is exhausted. We set $\phi(0) = 0$ and interpret $F(\cdot \mid 0)$ as degenerate at zero. The boundary problem therefore yields $V_m(b, 0) = b$.

Proposition D.3 (Existence, uniqueness, monotonicity, and Lipschitz property)**.** *For each $m \in [h_L, h_T]$, there exists a unique bounded value function V_m solving (D.5)–(D.6). Moreover:*

1. *$V_m(b, \bar{s}) \geq b$ for all $(b, \bar{s})$;*
2. *$V_m(b, \bar{s})$ is weakly increasing in b;*
3. *$V_m(b, \bar{s})$ is Lipschitz in b with constant one;*
4. *$V_m(b, \bar{s})$ is weakly increasing in $\bar{s}$.*

Proof. For bounded functions W_1, W_2,

$$|\max\{b, x, W_1(\max\{b, x\}, \phi(\bar{s}))\} - \max\{b, x, W_2(\max\{b, x\}, \phi(\bar{s}))\}| \leq \|W_1 - W_2\|_\infty.$$

Also,

$$|W_1(0, \bar{s}) - W_2(0, \bar{s})| \leq \|W_1 - W_2\|_\infty.$$

Hence

$$\|\mathcal{T}_m W_1 - \mathcal{T}_m W_2\|_\infty \leq \frac{\lambda + m}{\rho + \lambda + m} \|W_1 - W_2\|_\infty.$$

Since $\rho > 0$, the modulus $(\lambda + m)/(\rho + \lambda + m)$ is strictly below one. Therefore $\mathcal{T}_m$ is a contraction, and Banach's fixed-point theorem gives a unique bounded fixed point V_m.

The inequality $V_m(b, \bar{s}) \geq b$ follows immediately from the stopping branch in (D.7).

To show monotonicity in b, suppose W is weakly increasing in its first argument. For $b_2 > b_1$,

$$\max\{b_2, x, W(\max\{b_2, x\}, \phi(\bar{s}))\} \geq \max\{b_1, x, W(\max\{b_1, x\}, \phi(\bar{s}))\}$$

pointwise in x. Thus $\mathcal{T}_m W$ is weakly increasing in b. Starting from a weakly increasing function and iterating $\mathcal{T}_m$, the contraction limit V_m is weakly increasing in b.

For the Lipschitz property, suppose W is Lipschitz in b with constant one. Then for $b_2 > b_1$,

$$\max\{b_2, x, W(\max\{b_2, x\}, \phi(\bar{s}))\} - \max\{b_1, x, W(\max\{b_1, x\}, \phi(\bar{s}))\} \leq b_2 - b_1.$$

The continuation branch is therefore Lipschitz in b with constant at most $\lambda/(\rho + \lambda + m) < 1$, while the stopping branch b is Lipschitz with constant one. Hence $\mathcal{T}_m W$ is Lipschitz in b with constant one. Iteration and uniform convergence preserve this property.

Finally, suppose W is weakly increasing in $\bar{s}$. If $\bar{s}_2 > \bar{s}_1$, then $F(\cdot \mid \bar{s}_2)$ first-order stochastically dominates $F(\cdot \mid \bar{s}_1)$ and $\phi(\bar{s}_2) > \phi(\bar{s}_1)$. Since the integrand in (D.7) is weakly increasing in both x and $\phi(\bar{s})$, $\mathcal{T}_m W$ is weakly increasing in $\bar{s}$. Iterating from a weakly increasing function and using contraction convergence gives the result. □

Lemma D.1 (Continuity in the perceived hazard)**.** *For each $(b, \bar{s})$, $V_m(b, \bar{s})$ and $C_m(b, \bar{s})$ are continuous in $m \in [h_L, h_T]$. Consequently, if $C_m(b, \bar{s}) < b$, then there exists $\eta > 0$ such that $C_{m'}(b, \bar{s}) < b$ whenever $|m' - m| < \eta$.*

Proof. The operator $\mathcal{T}_m$ is continuous in m uniformly over bounded functions on the compact interval $[h_L, h_T]$. Moreover, its contraction modulus is uniformly bounded away from one:

$$\sup_{m \in [h_L, h_T]} \frac{\lambda + m}{\rho + \lambda + m} < 1.$$

The standard parametric contraction theorem therefore implies that the fixed point V_m is continuous in m under the sup norm. Since C_m is obtained from V_m and m through the continuous expression in (D.6), $C_m(b, \bar{s})$ is also continuous in m. The final statement follows immediately from continuity. □

E.4 The Stopping Threshold

Define the stopping set

$$S(m, \bar{s}) = \{b \in [0, \bar{x}] : V_m(b, \bar{s}) = b\}.$$

Lemma D.2 (Stopping set is an upper interval)**.** *For each $m \in [h_L, h_T]$ and $\bar{s} \in [0, \bar{s}_0]$, the stopping set is a nonempty closed upper interval:*

$$S(m, \bar{s}) = [b^*(m, \bar{s}), \bar{x}].$$

Moreover, the threshold $b^*(m, \bar{s})$ is weakly increasing in $\bar{s}$. Equivalently, the threshold weakly falls as the queue deteriorates.

Proof. Let

$$D_m(b, \bar{s}) = V_m(b, \bar{s}) - b.$$

By Proposition D.3, $D_m(b, \bar{s}) \geq 0$. For $b_2 > b_1$, the Lipschitz property gives

$$V_m(b_2, \bar{s}) - V_m(b_1, \bar{s}) \leq b_2 - b_1.$$

Therefore

$$D_m(b_2, \bar{s}) \leq D_m(b_1, \bar{s}),$$

so $D_m(\cdot, \bar{s})$ is weakly decreasing. Hence the set $\{b : D_m(b, \bar{s}) = 0\}$ is a closed upper interval.

The set is nonempty. Since payoffs are bounded above by $\bar{x}$ and immediate purchase of $\bar{x}$ is feasible, $V_m(\bar{x}, \bar{s}) = \bar{x}$, so $\bar{x} \in S(m, \bar{s})$. Define

$$b^*(m, \bar{s}) = \inf S(m, \bar{s}).$$

For monotonicity in $\bar{s}$, Proposition D.3 implies that $V_m(b, \bar{s})$ is weakly increasing in $\bar{s}$. Thus $D_m(b, \bar{s})$ is weakly increasing in $\bar{s}$. When the future queue is better, waiting is more valuable, the stopping set shrinks, and the threshold weakly rises. Hence $b^*(m, \bar{s})$ is weakly increasing in $\bar{s}$. □

A higher $\bar{s}$ means the buyer expects better future inspections, so she is more selective and requires a higher recalled value before stopping. As the search queue deteriorates, $\bar{s}$ falls and the stopping threshold falls.

Proposition D.4 (Strict value of recall under directed search)**.** *Fix $m \in [h_L, h_T]$ and $\bar{s} \in (0, \bar{s}_0]$. Let $0 < b_1 < b_2 \leq \bar{x}$, and suppose b_1 lies in the continuation region:*

$$V_m(b_1, \bar{s}) > b_1.$$

Then

$$V_m(b_2, \bar{s}) > V_m(b_1, \bar{s}).$$

Proof. We use a finite-arrival coupling argument. Because $\phi^K(\bar{s}) \to 0$, and because $b_1 > 0$, the boundary problem at $\bar{s} = 0$ implies that continuing is strictly worse than stopping at b_1. Hence, for K sufficiently large,

$$C_m(b_1, \phi^K(\bar{s})) < b_1. \tag{D.8}$$

By Lemma D.1, there exists $\eta > 0$ such that, whenever $|m' - m| < \eta$,

$$C_{m'}(b_1, \phi^K(\bar{s})) < b_1. \tag{D.9}$$

Equivalently, b_1 lies in the stopping region at queue position $\phi^K(\bar{s})$ for all perceived hazards m' sufficiently close to m.

Under no-sale learning, the perceived hazard evolves continuously downward from m at time zero. Therefore there exists $\varepsilon > 0$ such that, if no monitored sale occurs before time ε, then

$$|m_t - m| < \eta \qquad \text{for all } t \leq \varepsilon. \tag{D.10}$$

Consider two buyers, A and B, who face the same market primitives and start from the same $(m, \bar{s})$, except that buyer A has initial recalled surplus b_1 and buyer

B has initial recalled surplus $b_2 > b_1$. Couple the inspection arrival times, the inspected physical houses, the surplus draws, and the sale-shock processes of all subsequently inspected houses. Couple the survival of B's initial recalled house to the survival of A's initial recalled house; this coupling is distributionally valid because both initial recalled houses have the same sale hazard conditional on the market state.

Let T_K be the time of the K-th inspection arrival. Define the event

$$E_{K,\varepsilon} = \{T_K \leq \varepsilon\} \cap \{\text{no monitored sale occurs before } T_K\} \cap \{x_j < b_1 \text{ for } j = 1, \ldots, K\}. \tag{D.11}$$

This event has strictly positive probability. The Poisson process has positive probability of generating K arrivals before ε. Conditional on the market state and on $T_K \leq \varepsilon$, the probability of no monitored sale before T_K is positive because hazards are finite. Finally,

$$\Pr(x_j < b_1 \mid \phi^{j-1}(\bar{s})) = F(b_1 \mid \phi^{j-1}(\bar{s})) > 0$$

because $b_1 > 0$ and the density is strictly positive on $(0, \bar{x})$.

On $E_{K,\varepsilon}$, no inspected draw exceeds b_1, and the initial recalled house survives. Thus buyer A's recalled value remains b_1, and buyer B's recalled value remains b_2. By (D.10), the perceived hazard at time T_K remains within η of m. By (D.9), once the queue reaches $\phi^K(\bar{s})$, b_1 lies in the stopping set. Hence buyer A stops no later than the K-th inspection arrival and receives payoff b_1. If buyer B mimics buyer A's stopping rule, then on the same event buyer B receives payoff b_2. The discounted payoff difference on $E_{K,\varepsilon}$ is therefore strictly positive:

$$e^{-\rho\tau}(b_2 - b_1) > 0,$$

where $\tau \leq T_K$ is the common stopping time under the mimicking policy.

Outside $E_{K,\varepsilon}$, buyer B's recalled value weakly dominates buyer A's recalled value along the coupled path, so mimicking buyer A gives buyer B a payoff weakly higher than buyer A's payoff. Taking expectations, the mimicking policy yields buyer B a strictly higher expected payoff than buyer A obtains under her optimal policy. Since mimicking is feasible for buyer B,

$$V_m(b_2, \bar{s}) > V_m(b_1, \bar{s}).$$

□

Directed search is essential for Proposition D.4. In an i.i.d. arrival model, the distribution of future houses does not deteriorate with search progress. The buyer below the stopping threshold waits for a draw that dominates her recalled option, and the value function can be flat in b below the threshold. Directed search breaks this flatness: even if no new draw dominates the recalled option, the queue eventually deteriorates enough that the buyer may optimally stop on recall.

Proposition D.5 (Value is weakly decreasing in the perceived hazard)**.** *If $m' > m$, then*

$$V_{m'}(b, \bar{s}) \leq V_m(b, \bar{s})$$

for all $(b, \bar{s})$. Moreover, the inequality is strict at any state $(b, \bar{s})$ that lies in the continuation region under m' and satisfies

$$V_{m'}(b, \bar{s}) > V_{m'}(0, \bar{s}). \tag{D.12}$$

Proof. Let $W = V_{m'}$. We show that W is a subsolution of the m-problem. Define

$$A_W(b, \bar{s}) = -\kappa + \lambda \mathbb{E}_{x|\bar{s}} \left[\max\{b, x, W(\max\{b, x\}, \phi(\bar{s}))\} \right].$$

If $W(b, \bar{s}) = b$, then

$$(\mathcal{T}_m W)(b, \bar{s}) \geq b = W(b, \bar{s}).$$

If $W(b, \bar{s}) > b$, then W satisfies the continuation equation at m':

$$W(b, \bar{s}) = \frac{A_W(b, \bar{s}) + m' W(0, \bar{s})}{\rho + \lambda + m'}.$$

Using this identity,

$$\frac{A_W(b, \bar{s}) + m W(0, \bar{s})}{\rho + \lambda + m} - W(b, \bar{s}) = \frac{(m' - m) \left[W(b, \bar{s}) - W(0, \bar{s}) \right]}{\rho + \lambda + m}. \tag{D.13}$$

By Proposition D.3, $W(b, \bar{s}) \geq W(0, \bar{s})$. Therefore the right-hand side of (D.13) is weakly positive, so

$$W \leq \mathcal{T}_m W.$$

Iterating the monotone contraction $\mathcal{T}_m$ yields

$$W \leq \mathcal{T}_m W \leq \mathcal{T}_m^2 W \leq \cdots \to V_m.$$

Hence $V_{m'} \leq V_m$.

If (D.12) holds and the state lies in the continuation region under m', then $W(b, \bar{s}) > b$ and $W(b, \bar{s}) > W(0, \bar{s})$. The right-hand side of (D.13) is strictly positive, so

$$W(b, \bar{s}) < (\mathcal{T}_m W)(b, \bar{s}).$$

By monotonicity of the contraction and convergence to V_m, this implies

$$V_{m'}(b, \bar{s}) = W(b, \bar{s}) < V_m(b, \bar{s}).$$

□

A higher perceived hazard makes waiting less attractive because the recalled option is more fragile. The weak monotonicity result is global. Strictness requires that the recalled option have strictly positive value relative to the zero-recall state. Directed search provides conditions under which this strict inequality holds for positive recalled values in the continuation region.

Proposition D.6 (Threshold decreases in the perceived hazard)**.** *If $m' > m$, then*

$$b^*(m', \bar{s}) \leq b^*(m, \bar{s}).$$

Equivalently,

$$S(m, \bar{s}) \subseteq S(m', \bar{s}).$$

Moreover, if the old threshold is interior and the strict-crossing condition

$$C_{m'}(b^*(m, \bar{s}), \bar{s}) < b^*(m, \bar{s}) \tag{D.14}$$

holds, then the threshold strictly decreases:

$$b^*(m', \bar{s}) < b^*(m, \bar{s}).$$

Proof. Take any $b \in S(m, \bar{s})$. Then $V_m(b, \bar{s}) = b$. By Proposition D.5,

$$V_{m'}(b, \bar{s}) \leq V_m(b, \bar{s}) = b.$$

Since immediate purchase is always feasible,

$$V_{m'}(b, \bar{s}) \geq b.$$

Thus $V_{m'}(b, \bar{s}) = b$, so $b \in S(m', \bar{s})$. Therefore $S(m, \bar{s}) \subseteq S(m', \bar{s})$ and

$$b^*(m', \bar{s}) \leq b^*(m, \bar{s}).$$

For strictness, let $b^* = b^*(m, \bar{s})$ and suppose (D.14) holds. By continuity of the continuation value in b, there exists $\delta > 0$ such that

$$C_{m'}(b, \bar{s}) < b \qquad \text{for all } b \in [b^* - \delta, b^*].$$

Hence $V_{m'}(b, \bar{s}) = b$ on this interval, so

$$[b^* - \delta, b^*] \subseteq S(m', \bar{s}).$$

Therefore

$$b^*(m', \bar{s}) \leq b^* - \delta < b^* = b^*(m, \bar{s}).$$

□

The strict-crossing condition rules out knife-edge tangencies at the old threshold. Under this regularity condition, a sale-induced increase in m strictly lowers the reservation threshold.

E.5 Timing of Recall Purchases

Proposition D.7 (Recall purchases occur only at event times)**.** *Suppose the buyer does not purchase at time τ, and let τ' be the time of the next event, where an event is either a monitored sale shock or an inspection arrival. Then, almost surely, the buyer does not execute a recall purchase on the open interval (τ, τ'). Hence recall purchases occur only at sale-shock times or inspection-arrival times.*

Proof. On (τ, τ'), no inspection arrives, so $\bar{s}_t = \bar{s}_\tau$. No sale is observed, so by Proposition D.2, the perceived hazard weakly falls: $m_t \leq m_\tau$. The recall state b is also constant. Since the buyer does not stop at τ,

$$b < V_{m_\tau}(b, \bar{s}_\tau).$$

By Proposition D.5, a weakly lower hazard gives a weakly higher value:

$$V_{m_t}(b, \bar{s}_\tau) \geq V_{m_\tau}(b, \bar{s}_\tau) > b.$$

Thus b remains in the continuation region throughout (τ, τ'), so no recall purchase occurs before the next event. □

Two event types can trigger recall purchase. A sale shock raises m and lowers the threshold. An inspection arrival lowers $\bar{s}$ and may lower the threshold, while also delivering a new draw. Thus recall purchase can occur at sale shocks or inspection arrivals, but not spontaneously between events.

E.6 Purchase Probabilities and Hazards

Fix a pre-event state $(b, m, N, \bar{s})$.

Current-house purchase at an arrival. At an inspection arrival, the buyer draws $X \sim F(\cdot \mid \bar{s})$ and the queue advances to $\phi(\bar{s})$. She purchases the newly inspected house if it is at least as good as the recalled house and lies in the post-arrival stopping set:

$$P^{cur}(b, m, \bar{s}) = \Pr\{X \geq b,\ X \geq b^*(m, \phi(\bar{s}))\} = \Pr\{X \geq \max\{b, b^*(m, \phi(\bar{s}))\}\}. \tag{D.15}$$

Arrival-triggered recall purchase. At the same arrival, the buyer may reject the new draw but fall back on her recalled house because the queue has deteriorated. This probability is

$$P^{arr}_{rec}(b, m, \bar{s}) = \Pr\{X < b\}\, \mathbf{1}\{b \geq b^*(m, \phi(\bar{s}))\} = F(b \mid \bar{s})\, \mathbf{1}\{b \geq b^*(m, \phi(\bar{s}))\}. \tag{D.16}$$

Sale-shock recall purchase. At a monitored sale shock, the perceived hazard jumps from m to m^+. Here m^+ denotes the posterior hazard after observing a sale starting from prior hazard m. Conditional on a sale occurring, the sold house is uniformly distributed among the N monitored houses. Hence the best recalled option survives with probability $1 - 1/N$. Conditional on survival, recall purchase occurs if b belongs to the post-shock stopping set:

$$P_{rec}^{shock}(b, N, \bar{s}) = \left(1 - \frac{1}{N}\right) \mathbf{1}\{b \geq b^*(m^+, \bar{s})\}. \tag{D.17}$$

Total purchase hazard. Inspection arrivals occur at rate λ and monitored sales occur at rate Nm. Thus the instantaneous purchase hazard is

$$\Lambda(b, m, N, \bar{s}) = \lambda P^{cur}(b, m, \bar{s}) + \lambda P_{rec}^{arr}(b, m, \bar{s}) + NmP_{rec}^{shock}(b, N, \bar{s}). \tag{D.18}$$

Equivalently, the sale-shock recall component can be written as

$$(N-1)m\,\mathbf{1}\{b \geq b^*(m^+, \bar{s})\}.$$

The first two terms are arrival-driven channels: purchase of the newly inspected house and fallback onto recall after the queue deteriorates. The third term is the sale-shock-driven recall channel, which is the main object of the option-loss experiment.

E.7 The Effect of a Sale Shock

Proposition D.8 (Sale shocks induce immediate, less selective purchase)**.** *Fix a pre-shock state $(b, m^-, N, \bar{s})$, with prior*

$$\pi^- = \frac{m^- - h_L}{h_T - h_L} \in (0, 1).$$

Suppose a monitored sale shock occurs and updates the perceived hazard from m^- to m^+. Then:

1. $m^+ > m^-$.
2. $b^*(m^+, \bar{s}) \leq b^*(m^-, \bar{s})$, *with strict inequality under the strict-crossing condition in Proposition D.6.*
3. $V_{m^+}(b, \bar{s}) \leq V_{m^-}(b, \bar{s})$.
4. *A buyer with*

$$b \in [b^*(m^+, \bar{s}),\, b^*(m^-, \bar{s}))$$

would have continued before the shock but purchases immediately after the shock, provided the best recalled house survives.

5. *The recall-purchase probability at the sale shock is*

$$P_{rec}^{shock}(b, N, \bar{s}) = \left(1 - \frac{1}{N}\right) \mathbf{1}\{b \geq b^*(m^+, \bar{s})\}.$$

Proof. Part 1 follows from Proposition D.1. Part 2 follows from Proposition D.6. Part 3 follows from Proposition D.5. For Part 4, if

$$b \in [b^*(m^+, \bar{s}),\ b^*(m^-, \bar{s})),$$

then b lies below the pre-shock threshold and weakly above the post-shock threshold. Hence the buyer continues before the shock but stops after the shock, provided the best recalled option has not been sold. The survival probability is $1 - 1/N$. Part 5 is exactly (D.17). □

A monitored sale does not improve the buyer's opportunity set. Instead, it raises the perceived hazard of further option loss, lowers the value of waiting, and expands the stopping set. The buyer transacts earlier not because available houses become more attractive, but because waiting becomes less valuable.

E.8 Implications

E.8.1 Choice-set size

Proposition D.9 (Conditional amplification by choice-set size)**.** *Fix the pre-shock state $(b, m, \bar{s})$ and condition on observing a sale shock with post-shock hazard m^+. For a buyer with $b \geq b^*(m^+, \bar{s})$, the immediate recall-purchase probability at the shock is*

$$P_{rec}^{shock}(b, N, \bar{s}) = \left(1 - \frac{1}{N}\right) \mathbf{1}\{b \geq b^*(m^+, \bar{s})\},$$

which is strictly increasing in N.

Proof. Holding the stopping indicator fixed,

$$\frac{d}{dN}\left(1 - \frac{1}{N}\right) = \frac{1}{N^2} > 0.$$

□

The amplification works through the survival factor. When the buyer monitors more houses, an observed sale is less likely to have eliminated her best recalled option, so conditional on the same belief update she is more likely to convert the shock into an immediate recall purchase. This result provides a model-based rationale for the empirical interaction between option loss and choice-set size. It is a conditional prediction: the unconditional treatment effect may also depend on how $(b, m, \bar{s})$ is distributed across buyers with different N.

E.8.2 Visits

Let $I(b, m, \bar{s})$ denote the expected number of future inspections before purchase starting from state $(b, m, \bar{s})$.

Proposition D.10 (Sale shocks reduce future inspections for induced buyers)**.** *Consider a buyer with*

$$b \in [b^*(m^+, \bar{s}),\ b^*(m^-, \bar{s})).$$

If the best recalled house survives the sale shock, then the buyer stops immediately after the shock. Her remaining number of future inspections falls to zero. If the sale shock removes the best recalled house, the effect on future inspections is ambiguous.

Proof. For a buyer with

$$b \in [b^*(m^+, \bar{s}),\ b^*(m^-, \bar{s})),$$

Proposition D.8 implies that she would have continued before the shock but stops immediately after the shock if the best recalled house survives. Therefore her remaining number of future inspections becomes zero. Before the shock she was in the continuation region, so the expected number of future inspections was strictly positive. Hence the shock strictly reduces future inspections for these induced buyers.

If the sold house is the best recalled option, the state changes from b to 0. The loss of this fallback option can increase the need for future search, so the net effect on future inspections is ambiguous in that event. □

Thus the model predicts fewer new-property inspections through the induced-stopping channel. The sharp result applies to buyers whose recalled option survives and whose state is shifted into the stopping region by the sale shock. For other buyers, the effect on future visits is theoretically ambiguous because the signal channel lowers the threshold, while the mechanical loss of a valuable fallback option can increase the need for continued search.

E.8.3 Transaction prices

The model is written in terms of net surplus, not prices. Nevertheless, it provides a natural interpretation of the empirical association between option loss and higher transaction prices.

Suppose net surplus can be decomposed as

$$u = q - p,$$

where q is the buyer's value of the house and p is the transaction price. A sale shock raises the perceived hazard and lowers the stopping threshold from $b^*(m^-, \bar{s})$ to $b^*(m^+, \bar{s})$. Therefore the buyer becomes willing to accept a lower net surplus.

For a house of given quality q, accepting a lower net surplus corresponds to a willingness to pay a higher price, or equivalently to a weaker bargaining position.

This is an interpretation rather than a literal price theorem. Price is not a primitive in the model, and a structural price result would require modeling the joint quality-price distribution or an explicit bargaining protocol. The model delivers the sharp implication at the level of reservation surplus: option loss lowers the buyer's reservation surplus and induces earlier, less selective purchase.